

Communication and Runtime Engineering for Two-Party ResNet Inference in HPMPC

ABY2 with HE and silent-OT preprocessing

Technical report, HPMPC dbg branch

hpmc 97772ef, ConvTriple fa7b952, flexNN b1ce619

ImageNet all_opt (Section 9): hpmc 92a2c0e, ConvTriple 8d8f244, flexNN 417f12b; round 8: hpmc

GPU and multi-batch (Section 3.3): hpmc 88c800b, ConvTriple 1f5b71e, flexNN

TE0 / TE1, 64-bit rings (Sections 9 and 10): hpmc ef298d1, ConvTriple 2d0b448, flexNN

October 3, 2026

Abstract

We describe the optimizations and fixes that make HPMPC’s two-party ABY2 inference of ResNet50 fast, reproducible and correct across all of its protocol variants. The work targets both communication and runtime, in the preprocessing phase (homomorphic encryption for linear layers, silent OT for Boolean material) and in the online phase (round latency and per-round work). For one ImageNet image, preprocessing drops from 17.1 s to 4.3 s on two AMD EPYC 7543 hosts (3.3 s on EPYC 9354) and its triple traffic from 1112 to 587 MiB. The online phase drops from 2.4–4.3 s to 0.70 s. BatchNorm triples shrink from 5.7 GB to 0.19 GB of traffic per 10 CIFAR-10 images, and Beaver 3- and 4-tuples need half the OTs and about 45% less traffic. Evaluating several conv-triple chunks at once halves the CIFAR-10 conv phase, and in multi-batch mode the eight lanes of a register share one convolution, which cuts the conv phase per process from 8.6 to 3.9 s and the preprocessing traffic by half. On CIFAR-10 we evaluate 30 protocol variants (three MSB adders, five ReLU families, fused and unfused BatchNorm) in single-batch and 192-image multi-batch mode on two machine pairs. On the EPYC 9354 pair, preprocessing takes 1.3–2.7 s per 10 images and the online phase 0.35–0.50 s; 192 images take 7.8–19.5 s of preprocessing. Public weights, which need no linear-layer triples, and fully shared weight masks (`A_KNOWN=0`) work in every variant. With the AdamW-trained ResNet50, every variant classifies 62–72 of 100 images (plaintext 76) and 118–131 of 192 (61–68%; plaintext 71.9%). Along the way we fixed a thread race, two connection hangs, four protocol bugs that broke whole variant families, and the sources of run-to-run nondeterminism: all 60 threaded builds reproduced their serial reference bit for bit, and the final single-batch builds still give the same outputs. For the artifact’s 36 single-batch all_opt configurations on ImageNet, preprocessing gets 2.9–7.1× faster with secret weights and the online phase 2.0–3.0× (EPYC 9354). Their A2bits configurations were not correct as given; the A2B bake that fixes them commits every ReLU input’s mask before the preprocessing pass, for free behind convolutions and residual sums and through a mask-only forward with public weights, and an optional output repacking cuts the conv triples’ traffic by 40%. On a GPU, the packed pipeline’s evaluator produces the conv triples 2.1–2.3× faster than the CPU and 5.6–7.0× faster than the previous GPU path. With ferret’s extensions and the random OTs’ hashing on the GPU as well, bit for bit emp’s, preprocessing on one machine shared by both parties is 1.7–2.2× faster than on its CPU in single batch and 2.1–2.4× in multi-batch, and the online conv products build their column matrix on the GPU. Exact truncation (TE0, TE1) now runs in 2PC, fused into the ReLU without a message of its own, and classifies like the reduced-slack TS1. The 2PC configurations also run with 64-bit rings: BFV with $t = 2^{64}$ outside SEAL, the cut as narrow adders, the reshare bake, and OT-based arithmetic triples, at 1.7–2.1× the traffic and the plaintext model’s accuracy at $F = 12$; the narrow adders also fixed a 32-bit DReLU bug of the prefix adders under the A2B bake.

1 Introduction

Two-party inference with a preprocessing phase splits the cost of a secure neural network into two parts. An input-independent *preprocessing phase* produces correlated randomness: multiplication triples for the linear layers, generated with homomorphic encryption (HE), and Boolean triples and tuples for the nonlinear layers, generated with oblivious transfer (OT). The *online phase* then evaluates the network on masked values and needs only a few messages per layer. HPMPC implements this design with an ABY2-style protocol [1] (`PROTOCOL=4`) and generates its preprocessing material with the CHEETAH [2] HE and silent-OT [4] machinery (ConvTriple).

This report collects the engineering work that took this pipeline from a working prototype to a fast and reproducible system. Its focus is communication and runtime; Fig. 2 summarizes the largest effects.

- **HE preprocessing** (Section 3): a packed and pipelined convolution evaluator, an exact wire format that sends only the bits decryption needs, and BatchNorm triples packed into full ciphertexts and generated for all layers in one product, several chunks evaluated at once, and in multi-batch mode one convolution for all lanes. ImageNet preprocessing: $17.1\text{ s} \rightarrow 4.3\text{ s}$ ($4.0\times$).
- **Boolean preprocessing** (Section 4): OT packs sized by total demand, ferret parameters and a prefetching LPN step, TCP_NODELAY on every channel, bit-packed random OTs, and Beaver tuples whose products share one OT per common factor (half the OTs, $\approx 45\%$ less traffic).
- **Online phase** (Section 5): Nagle-free sockets, element-parallel circuit levels for batch size one, pooled GEMM threads, an ahead-of-time PRNG and fewer rounds per ReLU. ImageNet online: $2.4\text{--}4.3\text{ s} \rightarrow 0.70\text{ s}$.
- **Correctness and reproducibility** (Section 7): deterministic OT and HE randomness, and fixes for a worker race, lost wake-ups, stale listeners, multi-batch A2B, the threaded GEMM, and the BatchNorm of `A_KNOWN=0` and of public weights.
- **Evaluation** (Section 8): $30\text{ variants} \times 2\text{ batch modes} \times 2\text{ machine pairs}$, with final accuracy on a trained model.
- **The all_opt configurations on ImageNet** (Section 9): 36 builds over three weight settings, two ReLU protocols and three adders. The A2bits configurations as given compute wrong signs; the A2B bake fixes them by committing every ReLU input's mask before the preprocessing pass, at no traffic behind convolutions and residual sums. The conv triples start alongside the OT phase. Preprocessing $2.9\text{--}7.1\times$ faster with secret weights, online $2.0\text{--}3.0\times$.
- **Exact truncation** (Section 9): TE0 and TE1 fused into the ReLU on TS1's machinery, exact without slack; the low carry comes from a small adder in the ReLU's rounds (a four-way PPA of 2 rounds for prefix builds), its mask products from B2As instead of a multiplexer. TS1, TE1 and TE0 stated as protocols with proofs of the identities they compute and of their security (Section 9.1).
- **Generated a-known circuits** (Section 9): the circuit generator emits the a-known four-way PPA correctly at any width (pending a-known products in dot chains), gives a dot group's members one mask share, and folds the a-known RCA's first carry; A2bits PPA4 keeps the a-known circuit at the cut (368 MiB and $1.0\text{--}1.2\text{ s}$ less preprocessing on ImageNet), A2bits RCA saves 49 rounds.
- **64-bit rings** (Section 10): HE triples modulo 2^{64} , generated 64-bit and narrow adders, the cut and the reshare bake at 64 bits, arithmetic triples by HE with 64-bit shares ($6\times$ faster than Gilboa multiplication); $N = 4096, 8192$ and output repacking in LAN and WAN (Table 4: repacking 14–20% faster preprocessing in the WAN).
- **GPU** (Section 3.3): a GPU evaluator inside the packed conv pipeline, and ferret's LPN step, GGM trees and the random OTs' hashing on the GPU, all bit-identical to the CPU's. Conv triples $2.1\text{--}2.3\times$ faster; end to end, on one shared machine, preprocessing $1.7\text{--}2.2\times$ faster than on the CPU, $2.1\text{--}2.4\times$ in multi-batch; the online conv products on the GPU with the column matrix built there.

2 Setting

Protocol. ABY2 shares a value v as a public masked value $m = v + \lambda$ together with additive shares of the mask $\lambda = \lambda_0 + \lambda_1$. Every mask is fixed in preprocessing, so a linear layer is local except for one message that re-masks its output, and its preprocessing is a triple over the masks, e.g. $[\lambda_w \cdot \lambda_x]$ for a convolution (Fig. 1). With `A_KNOWN=1` the model owner P0 holds the weight masks and the data owner P1 the activation masks (the AB2 triple); with `A_KNOWN=0` both are shared (AB). We name the three weight settings by who knows the weights: *UC1*, known to none (secret-shared, `A_KNOWN=0`); *UC2*, known to one party, the model owner (`A_KNOWN=1`); *UC3*, known to all (public weights, `PUBLIC_WEIGHTS=1`, no linear-layer triples). A ReLU converts its input to Boolean shares (A2B), extracts the sign with an MSB adder and injects the bit back into the arithmetic domain (bit injection). The adder sets the number of online rounds per ReLU: the ripple-carry adder (RCA) needs 27 AND levels, the parallel-prefix adders PPA and PPA4 (4-input prefix gates, which use Beaver 3- and 4-tuples) need fewer levels but more AND gates.

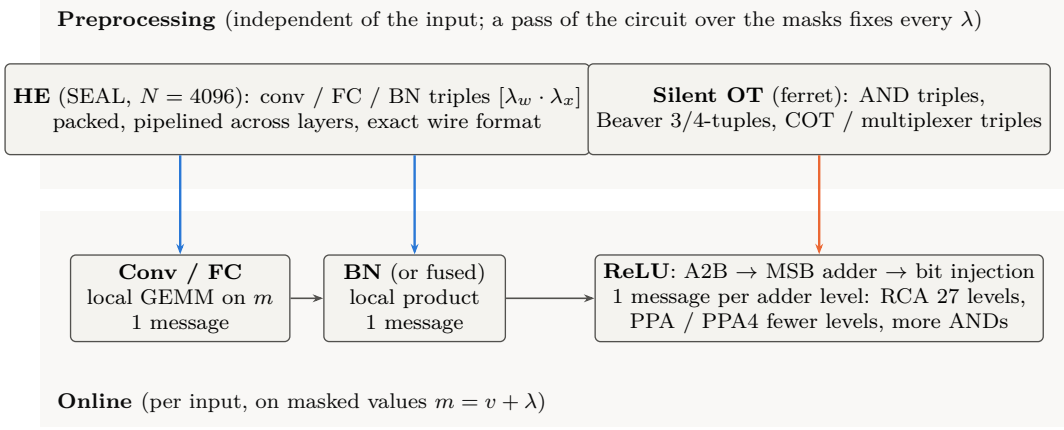


Figure 1: Where the cost of 2PC inference arises. A preprocessing pass fixes every mask; HE produces the linear-layer triples and silent OT the Boolean material. Online, a linear layer costs one message and a ReLU one message per level of its MSB adder.

ImageNet runs	flags beyond the <code>config.h</code> defaults
Model, input	<code>FUNCTION_IDENTIFIER=87</code> (default 15), <code>NUM_INPUTS=1</code> , <code>DATYPE=32</code> (one image, one process)
Preprocessing	<code>PRE=1</code> , <code>CHEETAH_CONV_PACKED=1</code> (from s1 on; s0 is CHEETAH’s HomConv2DSS), <code>CHEETAH_CONV_PIPELINE=1</code> (the default with packing, from s8 on), <code>CHEETAH_THREADS=32</code>
Network layers	<code>FUSE_CONV_BN=1</code> , <code>FUSE_RELU_AVG=1</code> , <code>RCA_MSB=1</code>
Roles	<code>MODELOWNER=P_0</code> , <code>DATAOWNER=P_1</code> (secret weights: <code>A_KNOWN=1</code> , <code>PUBLIC_WEIGHTS=0</code> , <code>TRUNC_DELAYED=0</code> and <code>MODELWEIGHTS_KNOWN_DURING_PREPROCESSING=0</code> were set explicitly, equal to the defaults)
ConvTriple (CMake)	<code>TRIPLE_ZERO=ON</code> ; <code>TRIPLE_FERRET=b12</code> from s10 on (b13 before)
Online steps (Fig. 10)	in addition <code>CUT_FRACTIONAL_BITS_OPT=1</code> , <code>SEND_BUFFER=100000</code> , <code>RECV_BUFFER=100000</code> , <code>ADDITIONAL_GEMM_THREADS=31</code> , <code>ADDITIONAL_RELU_THREADS=31</code> , <code>RNG_AHEAD=1</code>

Table 1: Build configuration of the ImageNet measurements. The other flags keep their defaults, e.g. `PROTOCOL=4`, `BITLENGTH=32`, `FRACTIONAL=5`.

Variants. We measure three adders (RCA, PPA, PPA4) and five ReLU families: *plain*; *reshare* (`RESHARE_OPT`); *reshare+sim*, which folds the reshare material into the convolution mask (`RESHARE_OPT_SIM`); *a2b*, which moves the A2B to preprocessing and bakes its mask into the convolution output (`A2B_ONLINE_OPT`, `A2B_CONV_BAKE`); and *a2b+AKTE*, an adder for inputs whose mask the evaluators know (`A_KNOWN_TO_EVALUATORS_OPT`). Each variant runs with BatchNorm fused into the preceding convolution or unfused. All variants share `BITLENGTH=32`, `FRACTIONAL=5`, probabilistic (SecureML-style) truncation, `FUSE_RELU_AVG=1` and `CUT_FRACTIONAL_BITS_OPT=1`.

Workloads. ResNet50 on CIFAR-10 in two modes. *Single batch*: one process, 32-bit lanes (`DATYPE=32`), 10 images, 32 HE/OT threads and 31 extra online threads. *Multi-batch*: 24 processes with 256-bit registers of eight 32-bit lanes, 192 images, 3 HE/OT threads and one extra online thread per process. The HE preprocessing study and the online steps use ResNet50 on one ImageNet image, `FUNCTION_IDENTIFIER=87`: CHEETAH’s ResNet50 layout for 1000 classes, whose 224×224 input arrives padded to 230×230 . It runs with a dummy model and image; since no code path depends on the values (Section 3.2), the times are those of a trained model. Table 1 lists every flag that differs from the defaults in `config.h`.

Measurement. Preprocessing and online time are P0’s wall-clock times (the maximum over processes in multi-batch). Single-batch numbers are medians of two runs; multi-batch runs once. All randomness is seeded, so every run of a build produces a bit-identical output; we check each run’s output hash against a serial build without extra threads (Section 7). The runtime does not depend on the weights: HE triples are generated with `TRIPLE_ZERO=ON` and no code path skips zero values. Accuracy uses two trained ResNet50 models: `Cifar_adam_001` for the runtime matrix and an AdamW-trained model (74.35% on the test set) for the final accuracy.

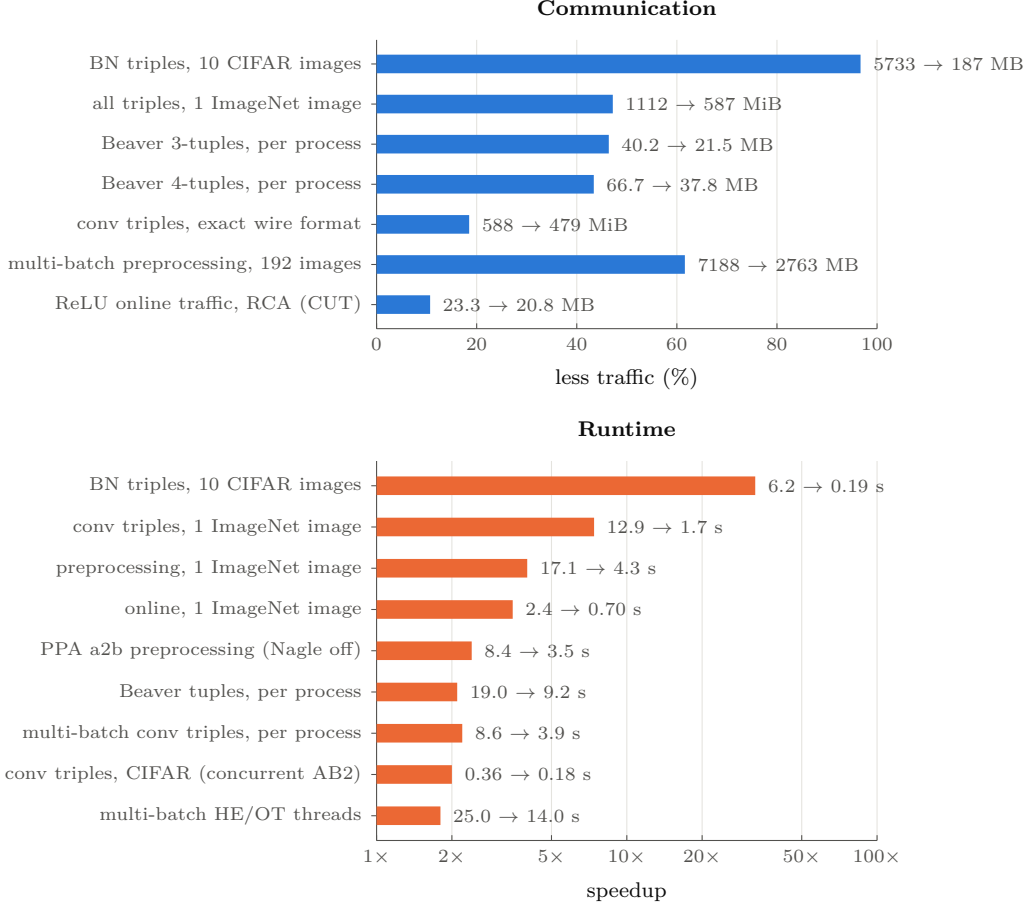


Figure 2: Headline effects of the changes in this report, each labeled with its before and after values. Communication: share of traffic removed. Runtime: speedup (log scale). Sections 3–6 explain each.

	flare / polynize	algofi / goracle
CPU	AMD EPYC 9354 (Zen 4), AVX-512, VAES, IFMA	AMD EPYC 7543 (Zen 3), AVX2, AES-NI
Cores, memory	32 cores / 64 threads, 752 GB	32 cores / 64 threads, 512 GB
Link	25 Gbit/s direct (Intel E810, ice 2.6.7), MTU 9700, 24.7 Gbit/s iperf3, RTT 0.28–0.31 ms	
Roles	P0 (model owner): flare / algofi; P1 (data owner): polynize / goracle	

Table 2: Machines. Each host builds its own binaries (`-march=native`).

3 Linear layers: HE preprocessing

3.1 Convolution triples: construction and security

This subsection defines the convolution triple, the packed encoding and the protocol that generates it, and argues correctness, security and cost. The engineering steps that follow (Fig. 4) change how fast the protocol runs, not what it sends.

Notation. All arithmetic shares live in $\mathbb{Z}_{2^{32}}$. BFV works over $R_t = \mathbb{Z}_t[X]/(X^N + 1)$ with $N = 4096$ and plaintext modulus $t = 2^{32}$, so a plaintext coefficient is exactly one ring element. The ciphertext modulus is $q = q_0 q_1$ with primes of 60 and 49 bits (109 bits, 128-bit security by the HE standard’s table), $\Delta = \lfloor q/t \rfloor$, s is a party’s ternary secret key and χ the error distribution. Enc_s denotes secret-key and Enc_{pk} public-key encryption. A convolution $y = W \star x$ is computed after the polyphase reduction of `ConvLayout`: a stride- σ , padded convolution is rewritten as a stride-1, unpadded convolution over σ^2 phase channels, which yields exactly the strided outputs. We therefore only treat stride-1 valid convolutions of B images of $C \times H \times W$ with M filters of $C \times k_h \times k_w$.

The triple, and why AB2 halves the traffic. With masked values $m_v = v + \lambda_v$, the ABY2 online step of a convolution is

$$m_y = m_W \star m_x - m_W \star \lambda_x - \lambda_W \star m_x + \Gamma + \lambda_y, \quad \Gamma = \lambda_W \star \lambda_x,$$

computed share-wise (P0 adds $m_W \star m_x$). The preprocessing must deliver additive shares of Γ , the *convolution triple*. Write $\lambda_v = \lambda_v^0 + \lambda_v^1$ for the parties' shares.

- **AB** (A_UNKNOWN=0). Both masks are shared, and $\Gamma = \lambda_W^0 \star \lambda_x^0 + \lambda_W^1 \star \lambda_x^1 + \lambda_W^0 \star \lambda_x^1 + \lambda_W^1 \star \lambda_x^0$. The first two terms are local; each cross term needs one secure product in which one party holds a filter and the other an input: two HE products per layer, one in each direction.
- **AB2** (A_UNKNOWN=1). The model owner P0 holds the whole weight mask ($\lambda_W^1 = 0$), and $\Gamma = \lambda_W \star (\lambda_x^0 + \lambda_x^1)$ is a *single* product: P1 encrypts λ_x^1 , P0 adds its own λ_x^0 inside the ciphertext and multiplies by its plaintext filters.

AB2 therefore needs half the ciphertexts, half the bytes and half the HE work of AB: 479 against 958 MiB for the ImageNet convolutions and 99 against 197 MiB for 10 CIFAR-10 images (Table 7), with the same protocol for each product. It is secure because it is ABY2's own input semantics: the party that inputs a value chooses that value's whole mask. The weights are P0's input. With **SHARE_PREP** P0 sets $\lambda_W = -W$, so the public masked weight is $m_W = 0$ and P1's share is 0: P1 learns nothing about W , and P0 learns only a mask of its own input. Nothing about x reaches P0, since λ_x^1 stays encrypted under P1's key. AB is needed only when no single party owns the weights (e.g. a secret-shared model); with the weights known in preprocessing (**MODELWEIGHTS_UNKNOWN_DURING_PREPROCESSING**) the same single product is used, with P1's share of Γ prescribed by its PRNG stream instead of drawn fresh, so no extra message is needed.

Packed encoding. The encoding follows the block layout of troy's **Conv2dHelper**. A tiling is (b, c_i, c_o, h, w) with $b c_i c_o h w \leq N$, $h \geq k_h$, $w \geq k_w$; a tile covers b images and an $h \times w$ window, and yields $y_h \times y_w = (h - k_h + 1) \times (w - k_w + 1)$ outputs per image and filter. For a tile τ (first image β_0 , first row r_0 , first column s_0) and an input group g of c_i channels, and an output group γ of c_o filters,

$$\begin{aligned} \pi_x^{\tau, g}(X) &= \sum_{\beta < b} \sum_{c < c_i} \sum_{i < h} \sum_{j < w} x[\beta_0 + \beta, gc_i + c, r_0 + i, s_0 + j] X^{(\beta c_i c_o + c) h w + i w + j}, \\ \pi_W^{\gamma, g}(X) &= \sum_{o < c_o} \sum_{c < c_i} \sum_{a < k_h} \sum_{d < k_w} W[\gamma c_o + o, gc_i + c, k_h - 1 - a, k_w - 1 - d] X^{(o c_i + c_i - 1 - c) h w + a w + d}. \end{aligned}$$

Lemma 1. For $\beta < b$, $o < c_o$, $i < y_h$, $j < y_w$ let $e(\beta, o, i, j) = (\beta c_i c_o + o c_i + c_i - 1) h w + (k_h - 1 + i) w + (k_w - 1 + j)$. Then $[\pi_x^{\tau, g} \cdot \pi_W^{\gamma, g}]_{e(\beta, o, i, j)} = \sum_{c < c_i} \sum_{a < k_h} \sum_{d < k_w} x[\beta_0 + \beta, gc_i + c, r_0 + i + a, s_0 + j + d] \cdot W[\gamma c_o + o, gc_i + c, a, d]$ in R_t , i.e. the input group's partial sum of output $(\beta_0 + \beta, \gamma c_o + o, r_0 + i, s_0 + j)$, and $\sum_g \pi_x^{\tau, g} \pi_W^{\gamma, g}$ holds the full output there.

Proof sketch. A product term of input (β, c, i', j') and flipped tap (o, c', a, d) has exponent $(\beta c_i c_o + o c_i + c_i - 1 + c - c') h w + (i' + a) w + (j' + d)$. (i) The column offset $j' + d \leq w + k_w - 2$ never reaches $w + k_w - 1 + j$, so no term carries from the previous row onto an output position, and the row offset is at most $(h + k_h - 2)w + w + k_w - 2$, so a carry from the previous block lands below $(k_h - 1)w + k_w - 1$, the smallest output offset. (ii) Within the target block $\beta c_i c_o + o c_i + c_i - 1$ only $c = c'$ contributes; $i' + a = k_h - 1 + i$ and $j' + d = k_w - 1 + j$ select exactly the taps of the valid convolution after the flip. (iii) Exponents reach N only for $\beta = b - 1$, $o = c_o - 1$, $c > c'$, i.e. blocks $\beta c_i c_o + (0 \dots c_i - 2)$; negacyclic reduction maps them to blocks $0 \dots c_i - 2$ (with a sign), and output blocks are $\equiv c_i - 1 \pmod{c_i}$, so no wrapped term reaches an output coefficient. $\square$

The remaining coefficients hold partial sums of other channel pairings; they are masked like the outputs and never decrypted. The conv-triple test (**cheetah_conv_triple_test**) checks every output exactly against a plaintext convolution, for the 23 CIFAR-10 layer shapes, the ImageNet layers and randomized shapes.

Tiling. A layer needs $T = \lceil B/b \rceil \lceil (H - k_h + 1)/y_h \rceil \lceil (W - k_w + 1)/y_w \rceil$ tiles, $G_{\text{in}} = \lceil C/c_i \rceil$ input ciphertexts and $G_{\text{out}} = \lceil M/c_o \rceil$ output ciphertexts per tile. The tiling minimizes the bytes on the wire (below),

$$\min_{b, h, w, c_o} T \cdot (G_{\text{in}} \cdot \text{in} + G_{\text{out}} \cdot \text{out}(b c_o y_h y_w)), \quad c_i = \min(\lfloor N/(b c_o h w) \rfloor, C),$$

where in and $\text{out}(u)$ are the sizes of an input ciphertext and of an output ciphertext with u used coefficients. CHEETAH's **HomConv2DSS** fixes $c_o = 1$: one output ciphertext per filter and tile. On ImageNet's 7×7 stage, such a ciphertext carries 49 useful coefficients out of 4096; the packed layout puts up to $\lfloor N/(b h w) \rfloor$ filters and several images into one output ciphertext.

Protocol $\Pi_{\text{conv}}^{\text{AB2}}$ (one layer; all tiles τ , input groups g , output groups γ)		
<i>Inputs:</i> P0: λ_W, λ_x^0 , P1's public key pk_1 . P1: λ_x^1 , secret key s_1 . <i>Outputs:</i> P0: R ; P1: $\Gamma - R$ with $\Gamma = \lambda_W \star (\lambda_x^0 + \lambda_x^1)$, R uniform.		
1. P1 encrypts.	For each (τ, g) : draw a seed ς , expand $a = \text{PRG}(\varsigma) \in R_q$, draw $e \leftarrow \chi$, set $c_0 = -as_1 + e + \Delta \pi_x^{\tau, g}(\lambda_x^1)$. Send (ς, c_0) : 16 bytes and $N(60 + 49)$ bits.	
2. P0 evaluates.	Expand a from ς directly in NTT form; add its own share, $c_0 += \Delta \pi_x^{\tau, g}(\lambda_x^0)$. For each γ : $ct_{\tau, \gamma} = \sum_g \pi_W^{\gamma, g}(\lambda_W) \cdot (c_0, a)_{\tau, g}$ (plaintext-ciphertext products, accumulated lazily in 128 bits).	
3. P0 re-randomizes and masks.	Flood: add a polynomial with uniform 64-bit coefficients to c_0 (CHEETAH's $\min(64, \lceil \log q \rceil - \lceil \log t \rceil - 2)$ bits), switch to modulus q_0 , add a fresh $\text{Enc}_{pk_1}(0)$. Draw $R_{\tau, \gamma} \leftarrow R_t$ uniformly (all N coefficients) and subtract $\Delta R_{\tau, \gamma}$. P0's output share of output (β, o, i, j) is $R_{\tau, \gamma}[e(\beta, o, i, j)]$.	
4. P0 sends only what decryption reads.	Truncate for decryption as CHEETAH does: at q_0 (60 bits) drop the low 26 bits of c_0 and the low 14 bits of c_1 , the widths below which bits only carry noise. Send c_1 (all N coefficients, 46 bits each) and c_0 only at the used positions E_τ (34 bits each).	
5. P1 decrypts	the coefficients E_τ : $\lfloor (c_0 + c_1 s_1)_e \cdot t / q_0 \rfloor$ for $e \in E_\tau$, its share of the outputs.	

Figure 3: The AB2 convolution-triple protocol (PackedConv2D). AB runs it twice, once in each direction, with P0's and P1's roles swapped for the second cross term.

ImageNet ResNet50, one image	triples, P0 sent + received (MiB)	factor
CHEETAH HomConv2DSS ($c_o = 1$), SEAL serialization (s0)	1,112	1.00
packed layout (s1)	698	0.63
+ exact wire format, byte-optimal tiling (s9, s10)	587	0.53
conv triples only, final: AB2 (A_KNOWN=1) / AB (A_KNOWN=0)	479 / 958	1 / 2

Table 3: Where the conv-triple traffic goes (UC2, 32-bit ring). The first three rows include the Boolean and FC material (108 MiB, unchanged); the last row compares the two triple types with the same protocol.

Security. Theorem 1. $\Pi_{\text{conv}}^{\text{AB2}}$ securely computes the functionality that outputs a uniformly random R to P0 and $\Gamma - R$ to P1 against a semi-honest adversary corrupting one party, assuming decision-RLWE for (N, q, χ) , the seed expansion modeled as a random oracle, and the statistical circuit privacy of CHEETAH's flooding with its parameters.

Proof sketch. Corrupted P0. Its view is the pairs (ς, c_0) . With $a = \text{PRG}(\varsigma)$ modeled as a random oracle, a is uniform and public, which is exactly the RLWE setting (SEAL's seeded ciphertexts rely on the same argument). A hybrid over the ciphertexts replaces each $c_0 = -as_1 + e + \Delta m$ by a uniform element; the simulator sends uniform (ς, c_0) . *Corrupted P1.* Its output $\Gamma - R$ is uniform because R is. Before truncation, the ciphertext P1 receives is, by the flooding and the fresh encryption of zero, statistically close to a fresh $\text{Enc}_{pk_1}(\Pi)$ of the plaintext $\Pi = P - R$, where P is the product polynomial. Since R covers all N coefficients, Π is uniform in R_t : its used coefficients equal P1's output and the others are independent and uniform. The simulator therefore samples Π uniform subject to P1's output on E_τ , encrypts it with flooding, and applies the same deterministic truncation and selection of c_0 coefficients. The changes of this work leave this argument intact: the layout only changes where values sit (the tiling depends on public shapes only), and the exact wire format and the partial c_0 are deterministic functions of the message CHEETAH would send, so they cannot reveal more. The bits that are not sent are either zero after truncation (checked at run time) or coefficients that decryption never reads. Finally, the evaluator's work does not depend on the weights' values (Section 3.2), so its running time reveals nothing about zero weights, which matters when the weights are known in preprocessing. $\square$

Communication. An input ciphertext costs $in = 16 + N(60 + 49)/8 = 55,824$ bytes, against 62.4 KB as a zstd-compressed SEAL seeded ciphertext; an output ciphertext costs $out(u) = \lceil 34u/8 \rceil + 46N/8 = 23,552 + 4.25u$ bytes. For each layer, the tiling minimizes $T(G_{\text{in}} in + G_{\text{out}} out)$; since an input costs about twice an output, it favors fewer inputs. Two effects add up on ImageNet (Table 3). Packing several filters and images into an output ciphertext removes most of the output ciphertexts of the late stages, and the exact wire format then cuts the remaining bytes. Together with AB2, the conv triples of a ResNet50 inference need 479 MiB, about a quarter of the roughly 2,000 MiB of AB with CHEETAH's layout (s0 with A_KNOWN=0: 2,115 MiB of triples in total).

What is left. A model of the layout search and the wire format (docs/paper/he_model.py) reproduces the measured 244.1 MiB of input and 234.9 MiB of output ciphertexts of the ImageNet convolutions. Dense packing in the same format would need 234.7 MiB, so about half of the traffic is unused coefficients. At 56×56 a channel

fills 3,136 of 4,096 coefficients (77%). At 14×14 and 7×7 the layout has to trade inputs against outputs: a polynomial holds c_i input channels and c_o output-channel blocks, so inputs use about $1/c_o$ and outputs about $1/c_i$ of their coefficients (10–48% and 4–24% in the chosen layouts). Stages 3 and 4 send 237 MiB where dense packing would need 67 MiB. Three cheap levers do not help. A per-layer ring dimension $N = 8192$ (same 109-bit modulus) is never cheaper, except on the first layer at equal cost. Fewer bits of the input ciphertexts would raise the product noise, which P1 knows and the 64-bit flooding must hide: with 12 bits of headroom below $\Delta = 2^{77}$, at most about 16 of 109 bits could go, paid for with more output bits. The output format already keeps only the bits decryption needs.

The remaining lever is *output repacking*: the evaluator packs the used coefficients of several sparse output ciphertexts into dense ones with ring automorphisms (a trace per ciphertext, or PackLWEs-style merging). That needs key switching, and key switching needs a special prime on top of the 109-bit data modulus ($60 + 49$ bits: 2^{32} plaintexts and 64-bit flooding). At $N = 4096$ the 128-bit security bound is already exhausted by those 109 bits, so repacking forces $N = 8192$ (with a 60-bit special prime, 169 of at most 218 bits). With the layout free to use dense inputs ($c_o = 1$) and outputs packed after the evaluation, the model gives 134 MiB of inputs and 108 MiB of outputs, 243 instead of 479 MiB per HE product, plus 4.7 MiB of Galois keys that P1 sends once (13 keys, SEAL serialization). Keeping today’s layouts and repacking only the outputs would give just 420 MiB. Over the whole preprocessing (`repack_estimate.py`, measured traffic of the other components) this removes 22–38% of the traffic in UC1, whose AB triples need both products (1,230–2,150 to 766–1,686 MiB), and 14–29% in UC2 (790–1,710 to 558–1,478 MiB), or 19–32% and 12–23% counting the online phase as well. UC3 has no linear-layer triples and gains nothing. The runtime cost is in the key switches. SEAL takes 583 μ s for one Galois automorphism at $N = 8192$ on the Zen 4 hosts (1,308 μ s on Zen 3), single-threaded. Calibrated against the measured conv phase, the evaluation itself stays about the same (0.64 against 0.71 s on Zen 4), and the repacking adds 1.1 s with a merge tree ($7.2 \cdot 10^4$ key switches) or 8.2 s with traces ($5.4 \cdot 10^5$) on 32 threads; on Zen 3, 1.7 or 12.4 s. So repacking would make an inference’s preprocessing about 1.0 s slower on Zen 4 (1.8 s on Zen 3) to save 236 MiB per HE product. That breaks even at about 1.9 Gbit/s (Zen 3: 1.1 Gbit/s): it pays off on WAN links and loses on the 25 Gbit/s testbed.

Output repacking, implemented. CHEETAH_CONV_REPACK=1 (round 4 of Section 9) packs with PackLWEs’ merge tree. The input polynomial holds C channel slots per position, channels fastest (C a power of two), and a weight polynomial one filter, channel c at $X^{C \cdot p - c}$ for kernel offset p ; the product then holds each output at a multiple of C , and every other coefficient mixes different channels, also where it wraps around. The evaluator merges the products of C filters of a tile level by level, $m = C/2, \dots, 1$: with the automorphism $\sigma : X \mapsto X^{N/m+1}$, which fixes X^{2mk} and negates $X^{m(2k+1)}$, the node $A + X^m B + \sigma(A - X^m B)$ doubles the used coefficients of A and B and cancels their other coefficients at the multiples of m ; after the last level filter r sits at residue r . The products are multiplied by $C^{-1} \bmod q$ first (q is odd), which the $\log_2 C$ doublings cancel exactly, so only the key-switching noise of the $C - 1$ automorphisms adds up, far below the flooding, which follows the packing (as does the mask, over all coefficients). P1 sends its Galois keys once (both parties for AB triples, 1.8 MB each). Measured on the 53 ImageNet convolutions (32 threads, all triples exact): the byte-optimal tilings take 148,622 automorphisms for 264 MiB and 6.3 s of HE; pricing each automorphism at 2 KB in the tiling search gives 26,000 for 285 MiB in 2.1 s, against 479 MiB in 0.71 s without repacking. The model’s 243 MiB assumed dense outputs; with C a power of two, the $49 \cdot 2^k$ positions of the feature maps fill only 77% of a polynomial. End to end, preprocessing traffic falls by 18–31% in UC1 and 11–24% in UC2, for 1.3–1.7 s more preprocessing on Zen 4 (break-even about 1.2 Gbit/s for UC2, 2.1 Gbit/s for UC1); the online phase is unchanged.

Ring dimension. Without repacking, the packed evaluator can also run in an $N = 8192$ ring with the same 109-bit modulus (CHEETAH_CONV_POLY_N=8192): twice the slots per ciphertext, so fewer and larger ciphertexts. The responses do not shrink with it: an output element’s share of a sparse output ciphertext costs about c_i times the coefficient width, and the tiling’s c_i grows with $\sqrt{N}$, so an ImageNet HE product sends 629 instead of 479 MiB (1.31 $\times$). We measured it on flare / polynize in the LAN and with a WAN shaped on both hosts (`shape_network_alt.sh`, 20 ms per direction and 200 Mbit/s on every interface, reset afterwards; Table 4). Plain $N = 8192$ loses in both: by up to 0.08 s in the LAN, and by 2.7–5.8 s in the WAN, where the conv phase is bandwidth-bound and takes 1.16–1.25 $\times$ as long. The ring that saves traffic is $N = 8192$ with output repacking (CHEETAH_CONV_REPACK=1, above): 285 instead of 479 MiB per HE product. In the WAN it shortens preprocessing by 14–20% (UC2 A2bits RCA 30.2 to 25.9 s, UC1 37.6 to 31.0 s, UC2 reshared 25.7 to 20.5 s), the conv phase by 32–36%, while the LAN pays 1.0–1.5 s for its automorphisms. The online phase does not depend on the packing; in the WAN it is round-bound (1,714–1,818 rounds of about 19 ms; the folded a-known RCA of Section 10 saves 49 of them). $N = 4096$ stays the LAN default; with links below the break-even of about 1.2 Gbit/s (UC2) repacking is the better choice.

build	packing	LAN pre s	WAN pre s	WAN conv s	WAN online s	conv MiB
UC1 A2bits RCA	$N = 4096$	3.38	37.57	27.00	32.44	958
	$N = 8192$	3.34	43.34	33.81	32.55	1,258
	repacked	4.42	31.01	17.32	32.69	570
UC2 A2bits RCA	$N = 4096$	3.20	30.16	16.87	32.24	479
	$N = 8192$	3.17	32.88	20.50	32.16	629
	repacked	4.19	25.93	11.52	32.16	285
UC2 reshared RCA	$N = 4096$	2.07	25.67	16.58	33.49	479
	$N = 8192$	2.15	28.54	19.27	33.45	629
	repacked	3.61	20.51	11.02	33.49	285

Table 4: Packing of the conv triples (ImageNet, dummy weights, flare / polynize, hpmc 8c0ff0f; WAN: 20 ms per direction, 200 Mbit/s; the three packings interleaved, median of 3 WAN runs, one LAN run; conv traffic sent and received by P0). Plain $N = 8192$: CHEETAH_CONV_POLY_N=8192; repacked: $N = 8192$ with a special prime, CHEETAH_CONV_REPACK=1.

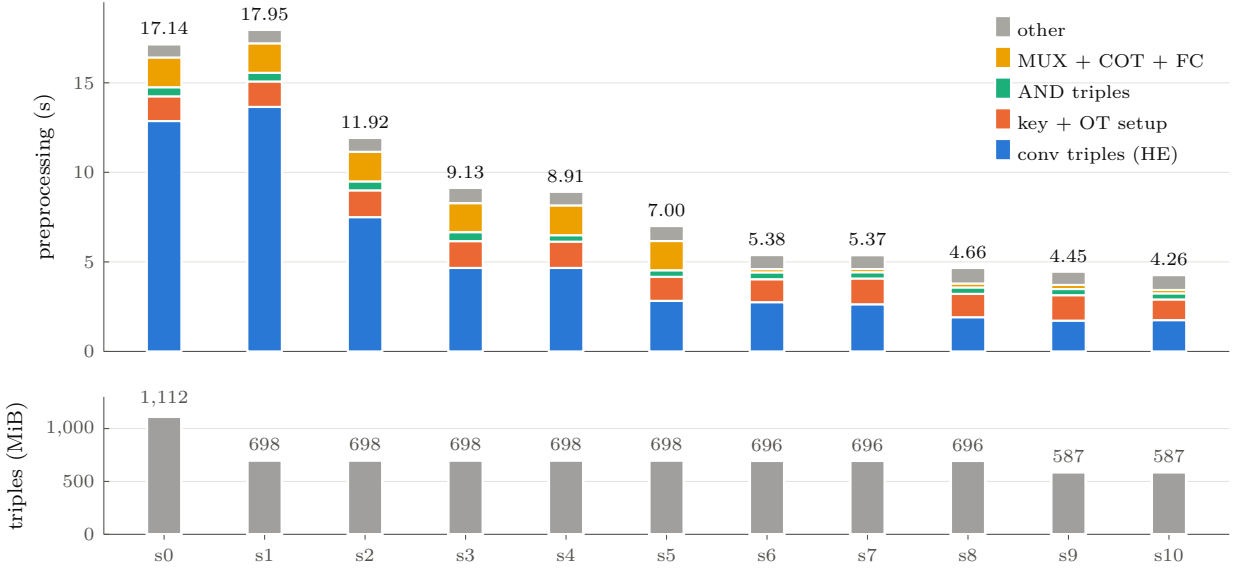


Figure 4: Preprocessing of one ImageNet image on algo / goracle, by component (median of 3 runs), and the triple traffic sent plus received by P0. s2–s6 are value-independent reruns (Section 3.2).

3.2 Engineering the HE preprocessing

Figure 4 traces ImageNet preprocessing through the ten ConvTriple changes (s1–s10); Table 5 lists what each step does.

The two machine pairs. We ran the s10 build on flare / polynize as well (Table 6). It sends exactly the same bytes, and preprocessing takes 3.31 instead of 4.26 s (22% less). Most of the gap is the convolution (1.13 against 1.74 s): the packed evaluator’s NTTs and multiply-accumulate use AVX-512 on Zen 4 and AVX2 on Zen 3. The ferret key setup and the AND triples, which are bound by memory and AES, gain only 12–14%. The online phase of this build was still dominated by delayed-ACK stalls on both pairs (medians 3.08 and 4.77 s); after the changes of Section 5 it takes 0.69–0.73 s on flare / polynize (1,820 rounds, of which P0 waited 0.15–0.16 s for data). The online steps were measured on flare / polynize only, and the PPA4 adder only there: preprocessing 10.8–11.4 s (before the shared-OT tuples) and online 1.05–1.06 s.

Packed evaluator (s1, s2). CHEETAH’s convolution encrypts one input tile per ciphertext and returns one ciphertext per output channel and tile. The packed layout of the GPU path puts several images and output channels into each ciphertext: triple traffic falls from 1112 to 698 MiB. At first this made the evaluator slower (s1), because every weight polynomial still took a full NTT. The evaluator now transforms a weight polynomial right before its only use, and it transforms 1×1 weights with NTTs of size $N/2^a$. It builds sparse polynomials from a table of powers and sums products lazily in 128 bits. Convolution time falls from 13.7 to 7.5 s.

Step	Change	Pre (s)	Conv (s)
s0	Baseline: CHEETAH’s HomConv2DSS, one output ciphertext per channel and tile	17.14	12.86
s1	Packed layout: several images and output channels per ciphertext (traffic 1112 $\rightarrow$ 698 MiB)	17.95	13.65
s2	Packed evaluator: weight polynomials transformed right before use, 1×1 weights through NTTs of size $N/2^a$, sparse polynomials from a power table, lazy 128-bit accumulation	11.92	7.49
s3	AES-CTR noise and masks instead of Blake2; ciphertexts loaded in parallel	9.13	4.66
s4	AND triples from bit-packed random OTs (64 per word)	8.91	4.66
s5	Thread pool kept across calls; multiply-accumulate blocked per prime	7.00	2.82
s6	Multiplexer and COT triples on all threads (MUX 1.08 $\rightarrow$ 0.07 s, COT 0.48 $\rightarrow$ 0.04 s)	5.38	2.74
s7	All work independent of the weights’ values (TRIPLE_ZERO=ON)	5.37	2.62
s8	Convolutions pipelined across layers (Fig. 5)	4.66	1.90
s9	Exact wire format and byte-optimal tiling (conv traffic 588 $\rightarrow$ 479 MiB)	4.45	1.71
s10	Prefetching LPN step, ferret parameters b12 (key setup 1.42 $\rightarrow$ 1.13 s)	4.26	1.74

Table 5: The HE and OT preprocessing steps (ImageNet, algoi / goracle, 32 threads). Table 6 compares the s10 build on both machine pairs.

ImageNet, one image, s10 build (median of 3 runs)	algoi / goracle (Zen 3)	flare / polynize (Zen 4)	ratio
Preprocessing (s)	4.26	3.31	1.29
key setup (ferret, HE keys)	1.15	1.01	1.13
conv triples	1.74	1.13	1.55
AND triples	0.34	0.29	1.17
multiplexer / COT / FC triples	0.07 / 0.03 / 0.10	0.08 / 0.07 / 0.08	
Triple traffic, P0 sent / received (MB)	305.5 / 281.9	305.5 / 281.9	1
Online (s), before the online changes	4.77	3.08	
Online (s), final online build (Fig. 10)	–	0.69–0.73	

Table 6: The same ImageNet build on both machine pairs. The online times of the s10 build vary with delayed-ACK stalls (flare / polynize 2.43–3.21 s over three runs).

Randomness and threads (s3–s6). Flooding noise and masks come from an AES-CTR generator (AES-NI) instead of SEAL’s Blake2; the thread pool survives across the several parallel calls of a layer; the multiplexer and COT triples run on all threads. Together these steps take preprocessing from 11.9 to 5.4 s.

Pipelining (s8). Per layer, each party idled while the other worked: at s7 P0 waited 0.82 s and P1 1.95 s in total. The pipelined protocol (Fig. 5) overlaps P1’s encryption and decryption with P0’s evaluation. It sends exactly the same bytes and cuts convolution time from 2.6 to 1.9 s; with A_KNOWN=0, where both parties encrypt, from 3.9–4.0 to 2.0 s in isolation.

Concurrent evaluation. Even pipelined, AB2 was slower than AB on CIFAR-10: 0.34–0.39 s against 0.26–0.30 s for the 23 convolutions of the isolated test, although in AB each party evaluates a full convolution under HE. Evaluating one chunk (up to 16 images) is a sequence of parallel regions: loading the inputs, the multiply-accumulate over blocks of output groups and primes, and masking, flooding and truncating the outputs. On CIFAR’s small layers these regions have fewer tasks than threads. An AB party keeps its threads busy by overlapping encryption, evaluation and decryption; the AB2 weight holder evaluated one chunk after another and sent each result before starting the next (Fig. 6). Now three evaluator threads take the chunks in order and a sender thread emits the outputs in chunk order. Chunk k draws its masks and flooding noise from the PRNG call that the serial loop gave it, so the triples are the same bits and every output hash is unchanged. Table 7 shows the effect of the number of chunks in flight. AB got the same change. End to end, the conv phase of a 10-image preprocessing halves, from 0.54–0.65 to 0.26–0.31 s.

Exact wire format (s9). SEAL serializes a seeded input ciphertext with zstd to 62.4 KB. P1 now sends only a 16-byte seed and c_0 at exactly $60 + 49 = 109$ bits per coefficient (55.8 KB). P0 expands c_1 from the seed directly in NTT form, which saves an inverse and a forward NTT per input. For outputs, the modulus switch before decryption keeps 34 bits of c_0 and 46 of c_1 ; exactly those bits travel, 20–25% less than zstd. The tile search now minimizes bytes rather than ciphertexts, since an input costs about 2.4 times an output. Convolution traffic falls 18% and all preprocessing traffic 12% (Table 8); decrypted values are bit-identical, and a check rejects outputs whose dropped bits are not zero. The 109-bit input modulus is a floor: BFV’s plaintext product carries a term of up to $\approx 2^{74}$.

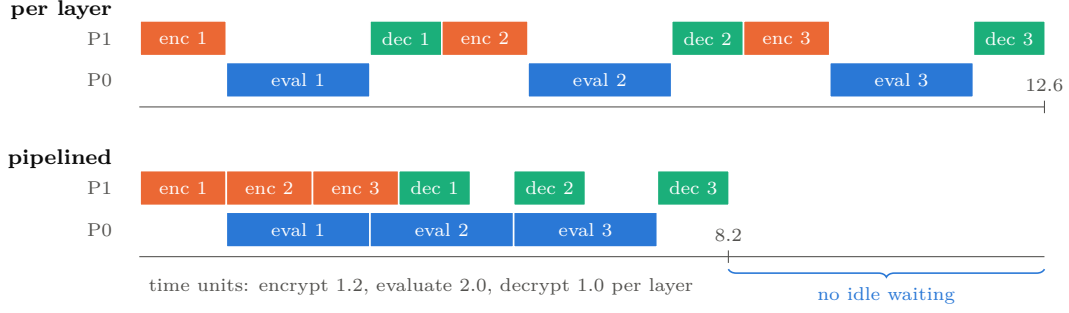


Figure 5: Pipelining the convolutions across layers (s8). Per layer, P1 waits while P0 evaluates and vice versa; pipelined, P1 encrypts the next layer’s inputs and decrypts the previous layer’s outputs while P0 evaluates. Same bytes, preprocessing 5.69 $\rightarrow$ 4.66 s.

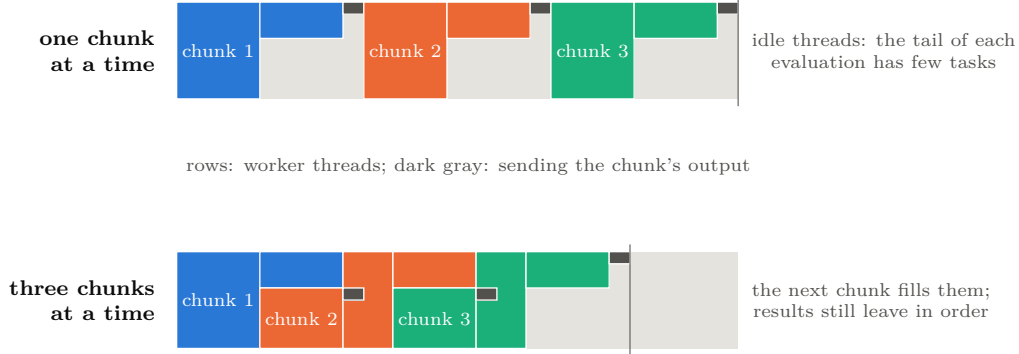


Figure 6: Why concurrent evaluation helps (schematic). The tail of each evaluation has fewer tasks than threads; with several chunks in flight, the next chunk fills the idle threads.

BatchNorm triples. Unfused BatchNorm multiplies each activation by a per-channel scale, which the slot-encoded protocol computes as an elementwise product of two vectors. The original code split each image’s elements among the 32 threads before packing: every thread sent its own nearly empty ciphertexts, one image at a time. Packing all images into full ciphertexts cut BN traffic from 2704 / 3029 to 90 / 101 MB and BN time from 6.2–7.9 to 0.7–0.8 s (CIFAR-10, 10 images). Each of the 53 BN layers was still a separate call, and one call is latency-bound (1.6–6.4 MB in 13–50 ms). `CHEETAH_BN_BATCHED` concatenates all slot-encoded layers into one product, split among the threads in whole ciphertexts: the BN phase falls from 0.75 to 0.19 s with the same traffic (Table 9).

Value independence. The benchmark’s dummy model makes every weight mask zero, and early versions of the packed evaluator skipped zero weights, so their benchmark times looked 0.6–2.9 s faster. With weights known in preprocessing, a zero-skipping evaluator would also leak the number of zero weights through its timing. All evaluators now do the same work for any weight values, and all numbers in this report are value-independent.

3.3 Conv triples and OT on a GPU

ConvTriple’s GPU path (`TRIPLE_GPU`, troy-nova) evaluates each convolution with troy’s `Conv2dHelper`, one round trip per layer, full ciphertexts on the wire. The packed CPU evaluator started from its tiling (s1) and then went further: the exact wire format, the pipeline across layers, concurrent evaluation, the start alongside the OT phase and the repacking variant. The GPU path got none of these. Now the packed pipeline itself has a GPU evaluator (`TRIPLE_GPU` builds, `CONV_GPU=0` turns it off), so all of them carry over. The device builds each weight polynomial directly in the shared memory of its NTT kernel (every coefficient, zero or a weight, computed from the weight tensor: no memset and no extra pass), transforms the inputs’ c_0 , sums the products with 128-bit accumulators (both halves of a ciphertext per thread, so each weight value is loaded once) and transforms the results back. The host keeps parsing the inputs, flooding, masking, truncation and the wire format, through pinned staging buffers. The transforms replicate SEAL’s `ntt_negacyclic_harvey` with SEAL’s own tables and slot order. So c_1 , which arrives in SEAL’s NTT form, needs no conversion, the triples are the same values as on the CPU, and the two-party check of all 53 ImageNet convolutions passes exactly.

isolated pipelined suites, flare / polynize, 32 threads chunks evaluated at once	AB2 (A_KNOWN=1)		AB (A_KNOWN=0)	
	CIFAR (s)	ImageNet (s)	CIFAR (s)	ImageNet (s)
1, sending in the evaluation loop (before)	0.34–0.39	–	0.26–0.30	1.03
1, separate sender thread	0.27–0.29	0.94		
2	0.19–0.20	0.77		
3 (default)	0.18–0.19	0.73	0.21	0.82
4	0.17–0.18	0.75		

Table 7: Conv triples with several chunks in flight (CONV_PIPE_EVALUATORS); every output checked exactly. Traffic is unchanged: CIFAR 99 MiB (AB2) and 197 MiB (AB), ImageNet 479 and 958 MiB.

MiB, P0 sent / received	before (s8)	after (s10)
Conv triples	232.6 / 355.1	234.9 / 244.1
Ferret setup	46.2 / 46.2	52.7 / 52.7
Multiplexer triples	35.4 / 35.4	35.4 / 35.4
COT triples	34.4 / 1.1	34.4 / 1.1
FC triples	0.7 / 1.2	0.7 / 1.2
Total	349.4 / 439.0 (788.4)	358.2 / 334.6 (692.8)

Table 8: Traffic of one ImageNet preprocessing (A_KNOWN=1). With A_KNOWN=0 the triple traffic falls from 659 / 626 to 551 / 518 MiB.

The conv triples alone. Table 10 compares CPU and GPU on one workstation, the only machine with a GPU, so both parties share its CPU. In isolation the GPU evaluator produces the conv triples $2.3\times$ (AB2) and $2.1\times$ (AB) faster than the packed CPU pipeline, and $5.6\text{--}7.0\times$ faster than the old GPU path, with 27% less traffic. In the whole inference that alone changes little (0.15–1.40s): on this machine preprocessing is the OT phase. With 16 one-channel OT packs per party, setting them up takes 2.0–2.5s, the A2bits builds’ Boolean addition (31 dependent COT rounds) another 4.8–5.1s, and the conv triples (4.6–6.4s on the CPU, 1.4–2.3s on the GPU) run alongside.

The OT phase on the GPU. Ferret’s extensions now run on the GPU as well, in four steps (Table 10, middle). (1) The LPN step: ten AES-indexed gathers per output, from a table of $k = 238,000$ blocks that stays in the GPU’s L2 cache; each lane keeps its own copy of the AES T-table in shared memory, so the table lookups never conflict. (2) MPCOT: all 2,507 GGM trees of an extension are built level by level, the sender’s from emp’s seeds, the receiver’s from its OT messages, with the per-level XOR sums reduced across warps. The LPN step then runs on the leaves in place, and one copy returns the outputs. Each channel’s tree messages leave as one message, the same bytes as emp’s 2,507 separate flushes. (3) The outputs stay on the GPU. In the ImageNet runs, 97% of all COTs go to random OTs with 1-bit messages (or k bit planes), which hash each COT with emp’s MITCCRH (a fresh AES key per OT) and keep one bit or a few. The GPU hashes them in place, scheduling each key on the fly, and returns only the bit planes: two to six bits per OT cross PCIe instead of 128, and the CPU does no hashing. The other consumers get host copies of their range on demand. (4) With extensions this cheap, setting up the packs dominated, so a GPU build uses 4 packs of 4 channels instead of 16 of 1: setup 0.8–1.2s, preprocessing 0.6–1.3s lower. Every step yields emp’s values bit for bit. Check modes compare every tree, output and hashed bit with emp’s CPU code. The Boolean triples pass the two-party tuple test, and seeded CIFAR-10 ResNet50 runs give the CPU build’s output hashes (with the same pack layout). The bytes on the wire do not change, so either party may run either path. Each ferret instance keeps its 164 MiB of outputs on the device; with both parties on one 20 GB GPU there is room.

End to end. With conv triples and OT on the GPU, preprocessing is $2.2\times$ faster than on the CPU in the A2bits builds and $1.7\text{--}1.8\times$ in the reshared ones (Table 10, bottom). What remains of preprocessing on the CPU is the packs’ setup (base OTs, IKNP and each pack’s first extension), the A2bits builds’ Boolean addition (latency-bound COT rounds), the 3% of COTs whose consumers need the blocks themselves, and the HE work around the GPU evaluator (encryption, flooding, decryption). HPMPC’s own CUTLASS convolution (USE_CUDA_GEMM=2) bypasses the per-output mask step and with it the A2B bake.

The online conv layers. A step timer over the online phase shows where a conv layer’s time goes (one ImageNet image, P0): 0.08s of 0.21s in the products, 0.05s in the per-output mask and send, 0.03s zeroing the outputs before the products, 0.03s adding the bias, 0.02s in im2col and 0.01s copying the bias masks for the A2B bake to every output. The zeroing and the bias add were serial loops and now run on the GEMM threads;

CIFAR-10, 10 images	traffic P0 sent / recv (MB)	BN phase (s)	preprocessing (s)
threads split each image	2704 / 3029	6.2–7.9	8.2–10.0
full ciphertexts	90 / 101	0.7–0.8	3.0–3.2
all 53 layers in one product	88 / 99	0.19	1.94 (RCA)

Table 9: BatchNorm triples with unfused BN. In multi-batch the per-process BN phase falls from 1.16 to 0.85 s (3 threads per process, compute-bound).

conv triples, ImageNet, 53 convolutions	AB2 (s)	AB (s)	traffic AB2 / AB (MiB)
old GPU path (troy, per layer)	5.26	10.22	655 / 1,310
CPU, packed and pipelined	2.18	3.02	479 / 958
GPU evaluator in the packed pipeline	0.94	1.45	479 / 958

ferret on the GPU: one OT pack, $2 \cdot 10^8$ Boolean triples	time (s)
CPU (emp-ot)	38.5–38.6
LPN step on the GPU	22.3–24.2
+ MPCOT’s trees on the GPU, LPN on the leaves in place	17.6–20.0
+ outputs copied straight into ferret’s buffer, one message per channel	14.4–15.1
+ outputs stay on the GPU, ROT bits hashed there	6.2–6.6

ImageNet, final flags	preprocessing (s)			online (s)		
build	CPU	GPU conv	GPU all	CPU	GPU conv	GPU all
UC1, A2bits, RCA	13.66	12.26	6.21	0.65	0.66	0.67
UC1, reshared, RCA	7.62	6.97	4.32	0.69	0.68	0.76
UC2, A2bits, RCA	12.56	11.86	5.73	0.46	0.48	0.53
UC2, reshared, RCA	6.73	6.58	3.89	0.51	0.51	0.56

Table 10: CPU against GPU on one workstation (RTX 4000 Ada, two Xeon Silver 4410Y with 24 cores, both parties on it, loopback). Top: the conv triples in isolation (16 threads per party, median of three runs), exact in every run. Middle: ferret’s extensions step by step (ConvTriple’s tuple test, both parties, two runs each). Bottom: the whole inference (hmpc 92a2c0e, ConvTriple 111e7be; 16 HE/OT threads and 11 pool workers per party, as the configurations’ 32 and 24 oversubscribe a machine both parties share; median of three runs). GPU conv: the conv triples on the GPU evaluator (LPN_GPU=0); GPU all: also ferret’s extensions and the ROT bits.

the bake takes one bias mask per channel. For the products, `GEMM_FAST_GPU` sent the `im2col`’d operand, nine times the input for a 3×3 kernel, over PCIe and gained nothing. Now the GPU builds the column matrix itself from the input feature map (only the input travels, and the host’s `im2col` is skipped), multiplies with the existing exact CUTLASS `uint32` GEMM, and returns the result. The output hashes are unchanged. With the parallel loops, the conv layers’ online time falls from about 0.21 to 0.15–0.18 s on the CPU and to 0.13–0.14 s with the GPU product, and the UC2 online phase of the GPU builds matches the CPU’s (0.425 against 0.426 s, Table 11). The UC1 builds (`A_KNOWN=0`) do not use `GEMM_FAST`; their online phase measured 0.04–0.05 s higher with the GPU OT path. That gap is still unexplained: pinning, a 5 s pause before the online phase, OT work during the online phase (there is none) and memory pressure are ruled out.

Multi-batch. In multi-batch mode (Section 6) a process holds 8 images in 256-bit registers. On the workstation, both parties’ processes share 62 GB of memory and one GPU, and other users’ jobs share them too. One ImageNet process per party peaks at 20–26 GB (GPU builds, both parties) and 26–30 GB (CPU builds). Two processes per party fit only for the GPU build, at 42 GB, and take 61.5 s for 16 images against 33.0 s for 8. Three do not fit. So Table 11 compares one process per party: preprocessing is 2.1–2.4× faster on the GPU, as in single batch, at 2.5–4.6 s per image against 3.9–6.0 s for single images (CPU: 5.5–11.2 against 7.1–13.3 s). The GPU is busy 90–100% of the time during the OT phase (Boolean triples and the bake’s Boolean addition, 22 s of the 33 s) while the CPU idles at a quarter: multi-batch preprocessing on this machine is bound by the OT kernels of two parties on one GPU. The conv triples, now one pipelined product over all lanes and running on channels of their own alongside the COT and multiplexer triples (Section 6), take 5.3 s on the GPU against 15.4 s on the CPU. The online phase gains less from the lanes: the activations scale linearly with the images (1.6–2.0 s for eight), and the UC2 conv layers need 0.77–1.19 s with the GPU product against 1.10–1.36 s on the CPU (1.5–1.7 s before the lanes’ `GEMM_FAST` and the parallel loops).

build	single batch, 1 image				multi-batch, 8 images (one process)			
	preprocessing (s)		online (s)		preprocessing (s)		online (s)	
	CPU	GPU	CPU	GPU	CPU	GPU	CPU	GPU
UC1, A2bits, RCA	13.29	6.04	0.59	0.64	89.59	36.80	4.14	3.97
UC1, reshared, RCA	7.73	4.33	0.64	0.68	50.92	24.69	4.32	4.40
UC2, A2bits, RCA	12.33	5.74	0.43	0.43	79.31	33.04	3.19	2.97
UC2, reshared, RCA	7.14	3.92	0.45	0.44	43.90	20.31	3.29	3.07

Table 11: CPU against GPU with the final code (hpmc 88c800b, ConvTriple 1f5b71e): GPU builds with the conv triples, ferret’s extensions and the ROT bits on the GPU and the online conv products there too (GEMM_FAST_GPU). ImageNet, both parties on the workstation; single batch 16 HE/OT threads and 11 workers per party, median of three runs; multi-batch DATTYPE=256 (8 images per process), one process per party with the same threads, median of two runs.

Protocol Π_{outer} (one element, direction $A \rightarrow B$; <code>cot_outer_multiply</code>) <i>Inputs:</i> A: $x_1^A, \dots, x_k^A$. B: y^B, or nothing if B’s share of y is to be fresh. <i>Outputs:</i> A: $u_1, \dots, u_k$; B: $v_1, \dots, v_k$ and, if fresh, y^B; with $u_p \oplus v_p = x_p^A y^B$. 1. $\mathcal{F}_{\text{ROT}}^k$: A gets $r_0, r_1 \in \{0, 1\}^k$, B gets s and r_s. 2. B corrects its choice. If y^B is given, B sends $e = s \oplus y^B$ (1 bit). If y^B is to be fresh, B sets $y^B := s$ and sends nothing; A uses $e = 0$. 3. A masks its k values with the sum of both messages: sends $m_p = x_p^A \oplus r_{0,p} \oplus r_{1,p}$ (k bits). 4. A outputs $u_p = r_{e,p}$. B outputs $v_p = r_{s,p} \oplus (y^B \wedge m_p)$. Both directions run in parallel, and each party adds its local term $x_p^P y^P$: shares of $x_p y$ for all p from two random OTs and, per party, k bits (plus one correction bit unless y is fresh).

Figure 7: Products $x_p y$ with a common factor from one random OT per direction.

4 Boolean material: OT preprocessing

4.1 Boolean correlations: protocol and security

All Boolean material is XOR-shared: $v = v^A \oplus v^B$ between the two OT roles A and B (P0 and P1, or reversed in the second direction). This subsection states the protocols of `ot_proto.hpp` and `silent_ot.h`.

Random OT with k -bit messages. The functionality $\mathcal{F}_{\text{ROT}}^k$ gives the sender two uniform strings $r_0, r_1 \in \{0, 1\}^k$ and the receiver a uniform choice bit s and r_s . Ferret silent OT (LPN parameters b12) produces correlated OTs: the sender gets $K_i \in \{0, 1\}^{128}$ and a global Δ with $\text{lsb}(\Delta) = 1$, the receiver $M_i = K_i \oplus s_i \Delta$ with $s_i = \text{lsb}(M_i)$. A tweakable correlation-robust hash H (emp’s MITCCRH, with a fresh key per call) turns them into random OTs, $r_b = H(i, K_i \oplus b\Delta)|_k$ for the sender and $r_{s_i} = H(i, M_i)|_k$ for the receiver. One hash yields 128 bits, of which `send/recv_rot_bitplanes` keep $k \leq 64$ as k independent message bits and store them as k bit planes, eight OTs per byte. By correlation robustness, r_{1-s_i} is pseudorandom to the receiver: a k -bit random OT costs one COT and one hash per message, the same as a 1-bit one.

Products with a common factor. Let $x_1, \dots, x_k$ and y be shared bits; the goal is shares of $z_p = x_p y$ for all p . Expanding, $z_p = x_p^A y^A \oplus x_p^B y^B \oplus x_p^A y^B \oplus x_p^B y^A$: two local terms and two cross terms, one in each direction. Figure 7 computes the cross term $x_p^A y^B$ for all p from a single $\mathcal{F}_{\text{ROT}}^k$ with A as sender. The other direction is symmetric, with B as sender.

Lemma 2 (correctness). $u_p \oplus v_p = x_p^A y^B$. *Proof.* If $y^B = 0$, then $e = s$ and $u_p \oplus v_p = r_{s,p} \oplus r_{s,p} = 0$. If $y^B = 1$, then $e = 1 - s$ and $u_p \oplus v_p = r_{1-s,p} \oplus r_{s,p} \oplus x_p^A \oplus r_{0,p} \oplus r_{1,p} = x_p^A$. With the fresh factor, $y^B = s$ and $e = 0$, and the same two cases hold. $\square$

Theorem 2 (security). In the $\mathcal{F}_{\text{ROT}}^k$ -hybrid model, Π_{outer} securely computes shares of $\{x_p y\}_p$ against a semi-honest adversary; with ferret and a correlation-robust hash realizing $\mathcal{F}_{\text{ROT}}^k$ (under the LPN assumption), so does the full protocol. *Proof sketch.* *Corrupted A (sender).* It receives $e = s \oplus y^B$; s is uniform and independent of A’s view, so e is a uniform bit and the simulator sends a uniform bit (nothing, if y^B is fresh). A’s output $u_p = r_{e,p}$ is uniform, and B’s output completes it to $x_p^A y^B$. *Corrupted B (receiver).* It receives $m_p = x_p^A \oplus r_{0,p} \oplus r_{1,p} = x_p^A \oplus r_{s,p} \oplus r_{1-s,p}$; since $r_{1-s,p}$ is uniform and unknown to B, m_p is uniform, and the simulator sends k uniform bits. With the fresh factor, B’s share $y^B = s$ is a uniform bit independent of A’s view, exactly as a freshly sampled share would be. All messages are simulated independently of the other party’s inputs, and every output share is uniform subject to the XOR relation. The k bits of one hash are

	random OTs		bits sent per party, products	
	one per product	Π_{outer}	one per product	Π_{outer}
3-tuple (3 products with c)	8	4	6	3
4-tuple (9 products of $\{a, b, ab\} \times \{c, d, cd\}$)	22	10	18	12

Table 12: Cost of one Beaver tuple, AND triples included in the OT counts. “One per product”: a separate correlated-OT multiplication for each product, as before this work.

independent pseudorandom bits under correlation robustness, so sharing one OT among k products does not correlate them. $\square$

Beaver tuples. A 3-tuple is (a, b, c) with shares of all products ab, ac, bc, abc ; a 4-tuple has (a, b, c, d) and all 11 products of two or more of them. PPA4’s 4-input AND gates consume them online in one round.

- **3-tuple:** an AND triple (a, b, ab) from two random OTs (`send/recv_rot_bits`), then one Π_{outer} per direction with $x = (a, b, ab)$ and the fresh $y = c$: ac, bc and abc . Total: 4 random OTs and, for the products, 3 bits per party.
- **4-tuple:** AND triples (a, b, ab) and (c, d, cd) (4 random OTs), then Π_{outer} with $x = (a, b, ab)$ for each $y \in \{c, d, cd\}$: the nine products $\{a, b, ab\} \times \{c, d, cd\}$ (6 random OTs). cd is not fresh, so these calls send the correction bit. Total: 10 random OTs and, for the products, 12 bits per party.
- **Party-local tuples** (`RESHARE_OPT_SIM`). P0 holds b and P1 holds c in full, the other shares being zero. The same protocol runs with the zero share as a given input: only P0’s zero share of c needs a correction bit.

Where the protocols are used. AND triples feed the RCA and PPA adders, 3- and 4-tuples the PPA4 adder, COT triples the bit injection and multiplexer triples the ReLU’s final selection. The A2B bake of the a2b family (`A2B_CONV_BAKE`) computes Boolean shares of $-(\ell_0 + \ell_1)$ for the committed conv output masks, where each party knows its own ℓ_i in full: a ripple-carry addition whose 31 carry levels are one multiplication each, run over all values of the network at once (ImageNet: 9.0 million values per level). Every OT consumer draws from the same OT packs (one per HE/OT thread, two ferret instances each, one per direction), which are sized from the total demand of the preprocessing before their first extension (`ot_demand_hint`) and extend in parallel.

4.2 Engineering the OT preprocessing

The ReLUs consume AND triples, Beaver 3- and 4-tuples (PPA4), COT triples (bit injection) and multiplexer triples. All come from ferret silent OT, 32 OT packs of two ferret instances each.

Ferret parameters and the LPN step. With 32 instances at once, ferret’s LPN step is memory-bound on its 7.2 MB tables. Computing the AES of the next four groups ahead and prefetching their table rows cuts the step from 0.34 to 0.28 s with bit-identical outputs. Parameter set b12 has a 3.8 MB table that stays in cache, for 6.5 MB more setup traffic per direction. Key setup falls from 1.42 to 1.13 s. VAES did not help: the LPN step is memory-bound and MITCCRH is bound by its AES key schedule.

Demand-sized OT packs. The first OT consumer used to size the packs, and later consumers extended them in small steps. `hpmc` now passes the total OT demand of the whole preprocessing to `ConvTriple` (`ot_demand_hint`) before the first generation. On CIFAR-10 the OT-pack phase falls from 0.91 to 0.47 s, and PPA4, reshare and a2b get enough capacity for their tuples.

TCP_NODELAY everywhere. The `ConvTriple` channels enabled `TCP_NODELAY` only in WAN mode. The COT multiply behind the a2b bake flushes small messages 31 times in sequence, and each flush waited for a delayed ACK. One call took 170 ms; with Nagle off it takes 11 ms. PPA a2b single-batch preprocessing falls from 8.4 to 3.5 s, PPA4 reshare+sim from 8.9 to 5.0 s. The same fix on `hpmc`’s own sockets takes the ImageNet online phase from 2.4–4.3 to 1.9–2.2 s (Section 5).

Bit-packed random OTs. Random OTs with one-bit messages are hashed and reduced to bits in cache-sized chunks and stored eight per byte (`send/recv_rot_bits`), where the generic interface kept two 16-byte blocks per OT. AND triples take 0.36 instead of 0.50 s, and the COT multiply reads the same bits with far less memory traffic.

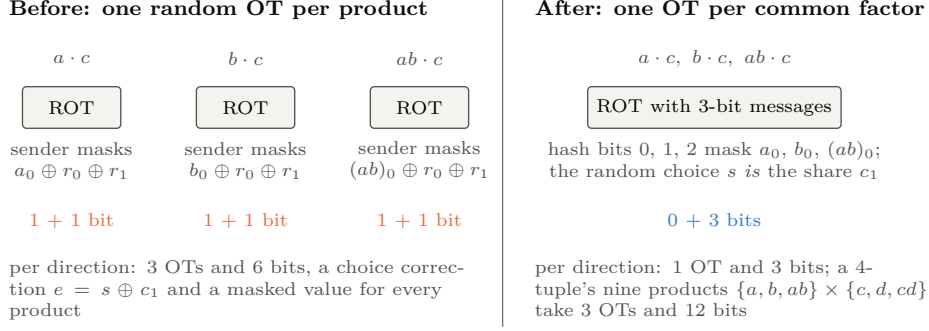


Figure 8: Products with a common factor share one random OT. Before, each product $x \cdot c$ took one random OT per direction plus two sent bits. Now one OT per direction carries all k products in its k message bits, and a party whose share of c is fresh takes it from the OT's random choice bit.

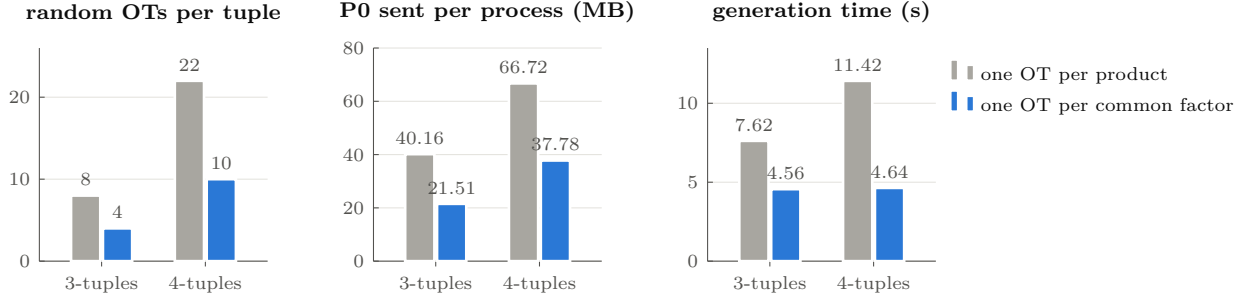


Figure 9: Beaver tuple generation per process in a multi-batch PPA4 run (flare / polynize, 24 processes).

Beaver 3- and 4-tuples with shared OTs. Before this work, a 3-tuple was an AND triple plus three independent correlated-OT multiplications $a \cdot c$, $b \cdot c$ and $ab \cdot c$, each with its own random OT per direction. Π_{outer} (Section 4.1) computes all products with the common factor c from one random OT per direction, whose k message bits (`send/recv_rot_bitplanes`) cost one hash, and takes the fresh c from the OT choices (Fig. 8). A 3-tuple now needs 4 random OTs instead of 8 and a 4-tuple 10 instead of 22 (Table 12). In a multi-batch PPA4 run the tuple phases fall from 19.0 to 9.2s per process and their traffic by 43–46% (Fig. 9). Unit tests check all products; ReLU exactness tests report 0 wrong outputs.

5 Online phase

On a 0.3ms LAN, the online phase is bound by round latency and per-round work, not by bandwidth. Activations send about 180 MB per ImageNet inference, 0.06 s at 25 Gbit/s. Figure 10 shows the steps for one ImageNet image.

Nagle, buffers and GEMM threads. With Nagle’s algorithm, a round’s last small message waited for a delayed ACK. ReLU layers were bimodal (6 or 46 ms), and the online phase varied between 2.4 and 4.3 s. With TCP_NODELAY it takes 1.9–2.2 s. Send and receive buffers of 100 000 words mean fewer messages and thread hand-offs per round. The threaded GEMM first sent its outputs in linear order while the completion visits 64×64 tiles; CIFAR-10 accuracy was 0% until both used the tile order. Its threads now live in a lock-free pool.

Element-parallel levels at batch size one. HPMPC vectorizes only across independent inputs, so a single image evaluated every level of a ReLU (adder steps, A2B, bit injection) element by element on one thread. Within a level, every element runs the same code and consumes the same number of words from each stream: sends, receives, preprocessed values, its own random words and the indices into the triple buffers. `stream_parallel_for` measures this consumption on the first element and runs the rest on the worker pool, each worker with cursors at exactly the stream positions the serial loop would use (Fig. 11). Segments end at send/receive buffer boundaries, and a misuse check verifies every worker’s consumption. The messages, their order and all values are those of the serial loop. With bulk random words, parallel adder construction and the GEMM mask-and-send and completion on the same pool, the ImageNet ReLU time falls from 630 to 300 ms and the online phase from 1.09 to 0.70 s. `RNG_AHEAD` moves the party’s own sequential AES stream to a background thread.

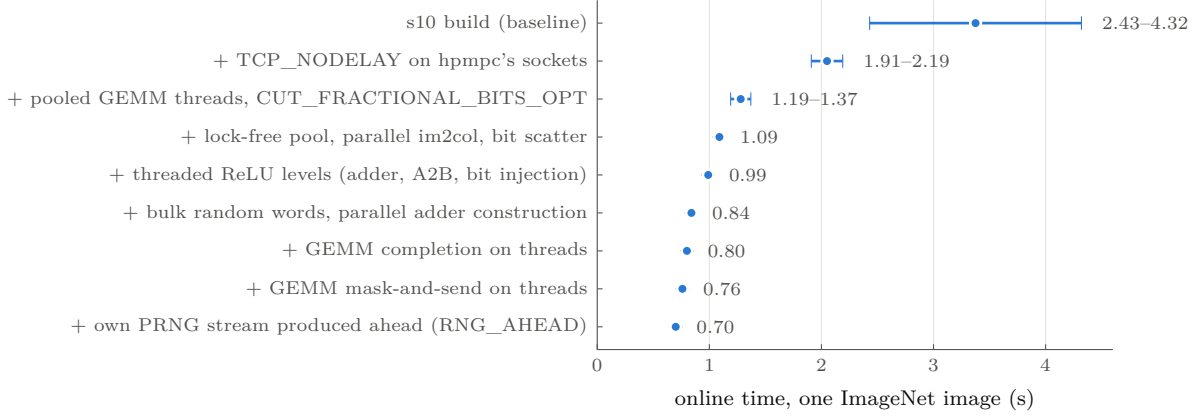


Figure 10: Online phase of one ImageNet image on flare / polynize (ranges over runs where they varied). Output hashes are unchanged by every step.

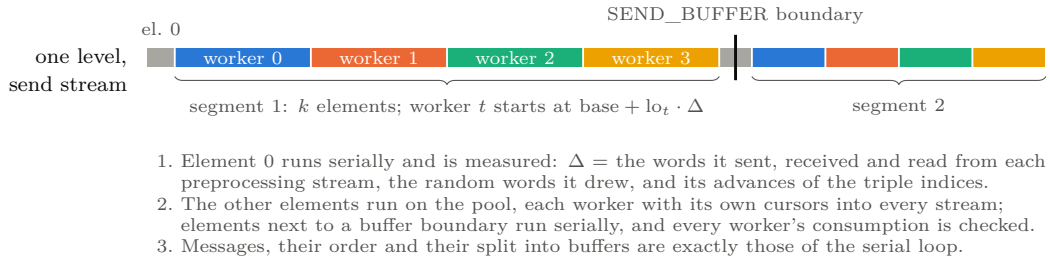


Figure 11: `stream_parallel_for`: element-parallel circuit levels at batch size one. One element is measured; the level's remaining elements run on the worker pool with per-worker cursors into every stream, split at buffer boundaries.

Rounds per ReLU. What remains is round latency: 1,752 rounds for CIFAR-10 with RCA against 776 with PPA and 625 with PPA4 (Table 13). `CUT_FRACTIONAL_BITS_OPT` removes the fractional bits from the MSB adder by substituting identity elements and public constants. RCA loses 245 of its 1,997 rounds (12%), and ReLU traffic falls 8–11% for every adder. Preprocessing and accuracy do not change measurably (1,134 vs. 1,138 of 1,752 evaluations correct with and without CUT).

6 Multi-batch

With 24 processes of 256-bit lanes, preprocessing dominates. With more than one HE/OT thread per process, the processes shared connections: channel i used port $base + i \cdot PROCESS_NUM$ while process bases stepped by 3, which caused hangs and wrong triples. With a stride of $3 \cdot PROCESS_NUM$, preprocessing falls from 25.0 to 14.0s with three threads per process (Table 15). One extra online thread per process cuts the online phase from 0.81 to 0.67s with an identical output hash.

Lanes in one convolution. Each 256-bit register holds eight images (lanes), and the conv triples were generated as one single-image convolution per lane and layer, each with its own copy of the weight masks: 424 small, unpipelined convolutions per process, with ciphertexts that are mostly empty. The model owner's mask of a weight is $-w$ (`SHARE_PREP`), and every lane's image uses the same weights, so all lanes have the same weight masks by construction. The eight triples of a layer are therefore one convolution over eight images with one set of weights. `CHEETAH_CONV_LANES` generates them that way and passes all layers to one pipelined call. A debug check confirmed that no lane's masks differ in any layer or process. With three HE/OT threads per process the conv phase falls from 8.4–8.7 to 3.8–3.9s and P0's preprocessing traffic by 50–62% (Table 14); four threads, which enable the pipeline, gain nothing more. With the AdamW model the accuracy is unchanged (PPA 121 of 192 against 123 without, RCA 131, PPA4 127). The premise does not depend on `A_KNOWN`: input sharing gives the model owner the share $-w$ and the other party 0 in every lane either way, and `A_KNOWN=0` only turns the triple into an AB triple, for which both parties encrypt their shares. The lane batching therefore also applies to `A_KNOWN=0`. In a local test (one process, 8 lanes of 4 images, 4 HE threads, AdamW model) the lane check stays silent for both parties, the conv phase falls from 12.9 to 7.2s and the conv traffic from 1036 to 540MB per direction, and 22 against 23 of 32 images are classified correctly. On flare / polynize (24 processes, 192 images) the conv phase per process falls from 9.7–10.0 to 4.1–4.4s and preprocessing by a factor

single batch, unfused BN	rounds		ReLU traffic (MB)		online (s)	
	CUT	no CUT	CUT	no CUT	CUT	no CUT
RCA	1,752	1,997	20.8	23.3	0.52	0.48
PPA	776	789	32.6	36.8	0.44	0.49
PPA4	625	625	20.1	21.8	0.42	0.42

Table 13: Online rounds, P0’s ReLU traffic and online time for 10 CIFAR-10 images on flare / polynize, with and without CUT_FRACTIONAL_BITS_OPT. The time differences are within run-to-run noise.

multi-batch, 192 images, flare / polynize	preprocessing (s)		conv phase per process (s)		P0 sent (MB) before → lanes
	before	lanes	before	lanes	
RCA, unfused	13.7	8.9	8.64	3.87	8,884 → 4,459
RCA, fused	12.5	8.0	8.43	3.86	7,188 → 2,763
PPA, unfused	16.9	12.0	8.73	3.88	9,022 → 4,597
PPA4, unfused	22.6	16.7	8.67	3.82	10,238 → 5,813

Table 14: Conv triples of all lanes as one convolution (CHEETAH_CONV_LANES), plain family, 3 HE/OT threads per process.

of 1.28–1.67 over the 30 variants (Section 8.1).

Weights known in preprocessing. The final ImageNet builds (Section 9) set MODELWEIGHTS_KNOWN_DURING_PREPROCESSING, and in multi-batch three of their single-batch optimizations were off. The lane batching excluded it: P1’s conv-triple shares are prescribed, and the lanes’ product now puts them in layer by layer after the batched product, as the single-batch pipeline does. Since the prescribed shares fix the triples, the outputs equal the per-layer path’s bit for bit. The A2B bake excluded it because its prescribed shares treated a register as one word; they are now computed lane by lane (Section 7). And the conv triples ran after the preprocessing pass, before the next generators. They now run on four channels of their own (CHEETAH_CONV_SIDE) alongside the Boolean addition, COT and multiplexer triples. The online products take all lanes as one GEMM with eight times the columns (GEMM_FAST for DATATYPE > 32: every lane holds the same weights, checked per layer). On the workstation (one ImageNet process per party, 8 images, UC2 A2bits RCA) the conv phase falls from 16.0 to 5.3s on the GPU and from 20.7 to 15.4s on the CPU, and preprocessing from 49.1 to 33.0s and from 86.8 to 79.3s (Section 3.3).

7 Correctness and reproducibility

Fast code is only useful if its outputs are right and repeatable. We made every run bit-reproducible, compared each threaded build against a serial one and bisected every configuration that lost accuracy over the added optimizations; this exposed the following bugs.

- **Nondeterministic OT.** COT and multiplexer triple shares are the ferret outputs, and hpmc truncates shares locally, so their per-run randomness changed the outputs. ConvTriple’s PRGs are now seeded per party, stage, call and task, and a scoped override seeds emp’s default PRGs per OT pack and worker (DetSeedScope).
- **Worker race in P0’s A2B.** For the reshare+sim skip with PPA4, P0 chooses each group’s slice masks from the Beaver 3-tuples that the group’s adder will consume, found through a count of groups prepared so far. On the worker pool that count was one shared global. Threaded PPA4 reshare+sim with fused BN gave a different output every run. The count is now a per-worker stream index. The race stayed hidden whenever the pool threads woke one after another.
- **Hangs.** A lost wake-up at connection start, where flag and broadcast were not under the waiters’ mutex, and a stale listener into whose backlog the next phase’s peer connected.
- **Multi-batch a2b.** The A2B mask bake derived the conv mask by casting transposed words to lanes, which holds only for 32-bit lanes; with 256-bit registers the a2b families classified 13–29 of 192 images. The mask is now computed value by value with the A2B’s own inverse packing, which leaves single-batch hashes unchanged, and the multi-batch a2b families classify 99–132 of 192 images, like the plain family.
- **Multi-batch a2b with weights known in preprocessing.** The bake still excluded MODELWEIGHTS_KNOWN_DURING_PREPROCESSING in multi-batch, so these builds ran the A2B without the bake, whose

HE/OT threads	preprocessing (s)	conv (s)	AND triples (s)	BN (s)	online (s)
1	25.0	16.5	3.6	2.1	0.63
2	17.0	10.2	3.3	1.4	0.68
3	14.0	8.8	3.0	1.2	0.75
4	14.0	7.8	3.8	1.0	0.69

Table 15: Multi-batch CIFAR-10 (24 processes, 192 images, RCA, unfused BN) on flare / polynize, by HE/OT threads per process, after the port-stride fix.

precomputed $[c]$ does not match the online mask: UC2 A2bits classified 2–4 of 32 CIFAR-10 images (the single-batch builds with the bake switched off likewise 0–2 of 10). P1’s prescribed share $-r_1 = (m_1 \ll F) + low$ is now computed lane by lane (the same values with one word per register: single-batch hashes unchanged), the bake covers these builds, and they classify 20 of 32 (UC1 reshared 21 of 32).

- **Threaded GEMM with A_KNOWN=0 or public weights.** The threaded GEMM had diverged from the serial one. With A_KNOWN=0 it sent the outputs without the conv triple’s share and left the triple cursor behind, so every threaded build classified at chance (1 of 10). With public weights and TRUNC_DELAYED=0 it truncated the first layer with the regular truncation while the completion expected the data owner’s exact one, and both parties waited forever in preprocessing. It now follows the serial path.
- **BatchNorm triples with A_KNOWN=0.** AB triples multiply the full masks, and the BN product’s operand is $x - \mu$, whose mask includes the model owner’s mask of μ . The triple was set up over the masks of x , which left out $\lambda_s \lambda_\mu = s\mu$: every output was $s \cdot x + \beta$ (unit test failed, ResNet50 unfused 0 of 10). With A_KNOWN=1 only the data owner’s mask share enters, and μ does not change it.
- **Public weights with unfused BN and delayed truncation.** With public weights the first convolution’s output is still the data owner’s sharing (the other party’s mask is zero). With TRUNC_DELAYED=1, with or without BIT_INJECTION_TRUNC_SIM, the BN truncated it with the SecureML local truncation, which wraps for positive values (0–1 of 10 correct). The owner now truncates it exactly and re-masks, in the same single message.
- **Test harness.** The ReLU tests fed unbaked inputs to the reshare+sim skip (35% wrong); the harness now declares its inputs unbaked.

After these fixes, all 60 threaded variants (30 single-batch, 30 multi-batch) reproduced their serial build’s output hash on flare / polynize (round 4), and repeated runs are bit-identical. The concurrent conv-triple evaluation keeps every single-batch hash of round 4. The lane batching gives the multi-batch triples different random shares, so those hashes changed; its correctness rests on the lane check and on the accuracy (Section 6 and Fig. 15). Hashes differ between the machine pairs, because the PRNG draws 512-bit AES blocks with VAES-512 on Zen 4 and 256-bit blocks on Zen 3.

8 Evaluation of all variants

Figure 12 shows preprocessing and online time for every variant on both machine pairs.

Single batch. On flare / polynize, preprocessing takes 1.3–1.8 s with RCA, 1.9–2.2 s with PPA and 2.1–2.5 s with PPA4 when BN is fused, and 1.6–2.0, 2.2–2.4 and 2.3–2.7 s unfused; the a2b families sit at the upper end. The online phase takes 0.35–0.50 s. Because it is round-latency-bound, PPA and PPA4 are slightly faster online than RCA despite more AND gates. The fastest configuration in total is RCA with fused BN (1.32 s + 0.39 s).

Multi-batch. 192 images take 7.8–11.7 s (RCA), 11.0–15.0 s (PPA) and 13.9–19.5 s (PPA4) of preprocessing and 0.54–0.81 s online. Per image that is 41–101 ms of preprocessing, against 132–267 ms in single batch. The a2b+AKTE family is the fastest PPA4 variant here (13.9–14.8 s): its adder needs fewer tuples.

Machines. In single batch, algo / goracle needs 1.13–1.25 $\times$ the preprocessing time of flare / polynize and 0.84–1.30 $\times$ the online time. Multi-batch is compute-bound, and there Zen 3 needs 1.46–1.75 $\times$ as long for preprocessing and 1.04–1.84 $\times$ online; it lacks AVX-512 for the HE NTTs and the GEMM. The lane batching gains less there (conv phase per process 14.8 $\rightarrow$ 8.0 s against 8.4 $\rightarrow$ 3.8 s; preprocessing 20–35% less against 19–38%). ImageNet shows the same pattern (Table 6).

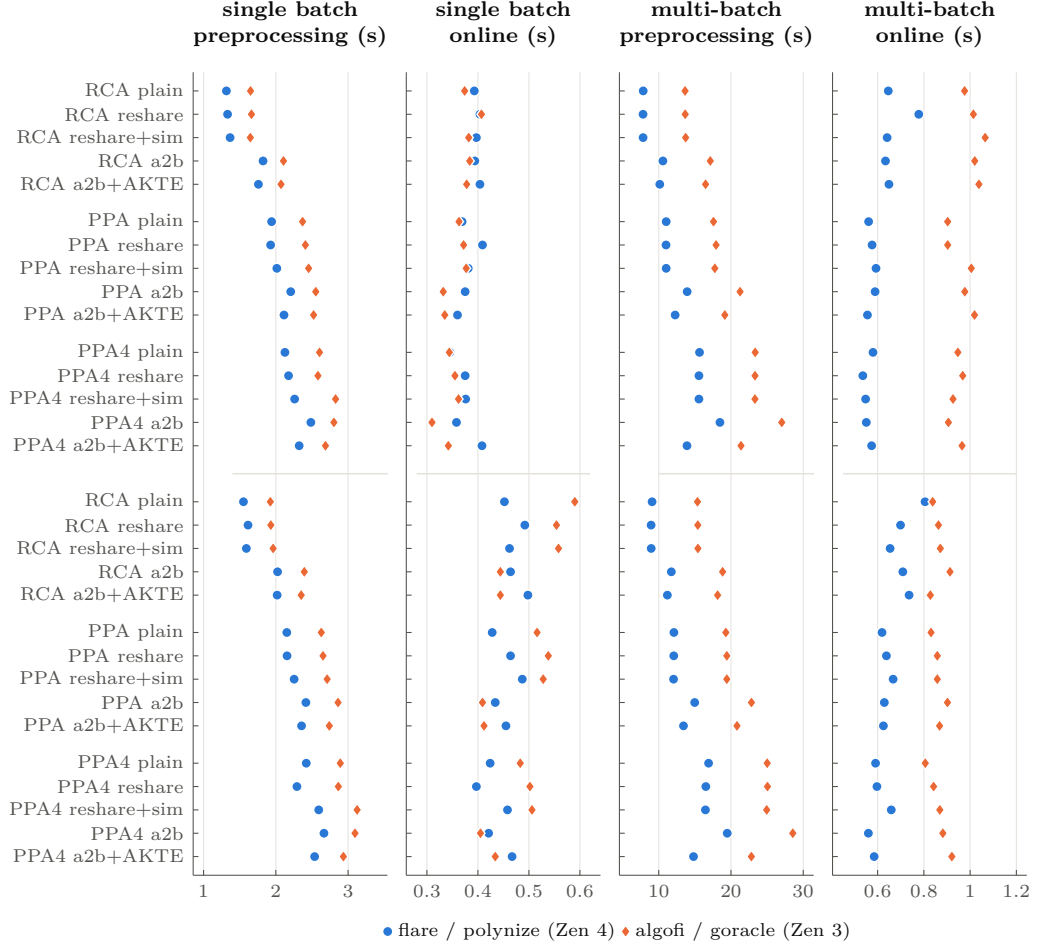


Figure 12: All 30 variants, CIFAR-10 ResNet50 (final builds). Upper block: fused BN; lower block: unfused BN. Single batch: 10 images, median of two runs. Multi-batch: 192 images, 24 processes. Tables with every value: [docs/VARIANT_PERFORMANCE.md](#).

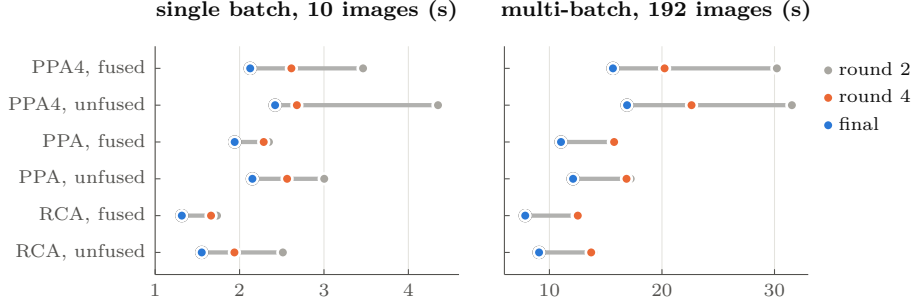


Figure 13: Preprocessing of the plain family in round 2, round 4 (after the shared-OT tuples and the batched BN triples) and with the final code (concurrent conv-triple evaluation; multi-batch: lane batching), flare / polynize.

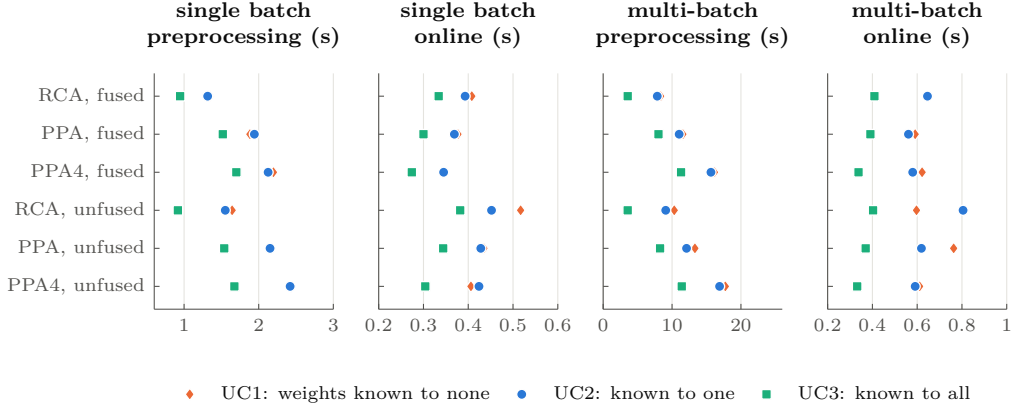


Figure 14: The three use cases, plain family, flare / polynize: UC1 (weights known to none, $A_KNOWN=0$), UC2 (known to the model owner, $A_KNOWN=1$, default) and UC3 (public weights, no conv triples). Final builds; multi-batch with lane-batched conv triples in UC1 and UC2.

The last optimizations. Figure 13 separates the steps. The shared-OT tuples and the batched BN triples (round 2 to round 4) help PPA4 most: single batch $4.35 \rightarrow 2.68$ s unfused and $3.46 \rightarrow 2.61$ s fused, multi-batch $31.5 \rightarrow 22.6$ s and $30.2 \rightarrow 20.2$ s; unfused RCA gains $2.51 \rightarrow 1.94$ s from the batched BN. The concurrent conv-triple evaluation then saves 0.26–0.55 s in every single-batch variant (10–26%), and the lane batching 4.5–6.3 s in every multi-batch variant (19–38%). From round 2 to the final code, preprocessing of the plain family falls by 17–44% in single batch and by 30–48% in multi-batch.

8.1 Weights known to none (UC1) and to all (UC3)

Besides UC2, the default, two use cases change who holds the weights. In UC1 ($A_KNOWN=0$) both parties hold shares of the weight masks as well as of the activation masks, so the conv, FC and BN triples are AB triples for which both parties encrypt. In UC3 (public weights, $PUBLIC_WEIGHTS=1$) the linear layers multiply locally and need no triples at all. We run public weights with $TRUNC_DELAYED=1$ and $BIT_INJECTION_TRUNC_SIM=1$: each linear layer’s truncation is deferred and folded into the next ReLU’s bit injection instead of costing a round of its own. Figure 14 compares the three use cases for the plain family.

UC3: public weights. Only the ReLU material remains to be preprocessed: across all 30 variants, preprocessing takes 0.92–2.03 s per 10 images and 3.5–14.3 s per 192 images, 16–41% and 22–61% less than the same variant in UC2 in the final builds. The online phase takes 0.27–0.41 s and 0.33–0.42 s (9–33% and 30–50% less): convolutions send nothing, and the truncation shares the bit injection’s message. With the AdamW model, the UC3 variants classify 63–73 of 100 images in single batch and 115–131 of 192 (60–68%) in multi-batch. Unfused BatchNorm needed the first-layer truncation fix of Section 7; before it, those variants classified 0 of 10 images.

UC1: weights known to none. With $A_KNOWN=0$ the linear-layer triples are AB triples, and the preprocessing traffic grows by a factor of 1.4–1.9 (RCA, unfused BN, 10 images: 229 to 428 MB). In single batch, preprocessing takes 1.34–2.74 s over the 30 variants and the online phase 0.34–0.52 s. Since the concurrent evaluation (Section 3), $A_KNOWN=1$ is faster in both HE phases of every variant (conv 0.27–0.33 against 0.30–0.39 s, BN

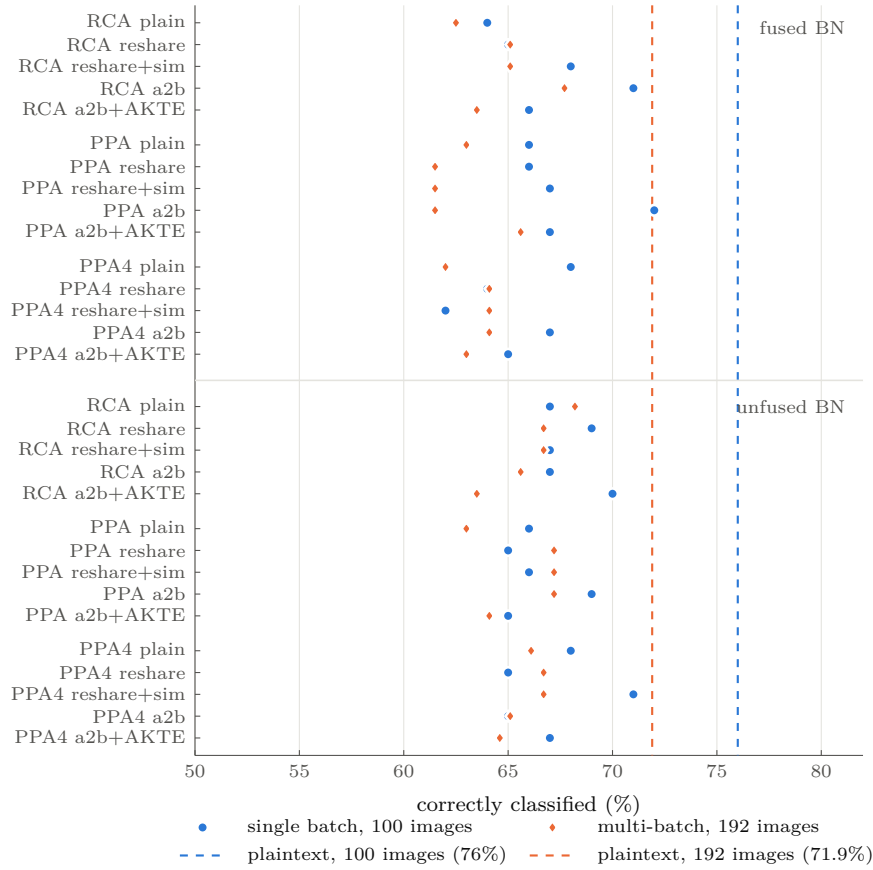


Figure 15: Accuracy with the AdamW-trained ResNet50, standard probabilistic truncation, final builds on flare / polynize: 100 images in single batch, 192 in multi-batch (lane-batched conv triples).

0.20–0.24 against 0.23–0.26 s); the totals differ by -0.07 to $+0.13$ s, within the noise of the OT phases. In multi-batch, with the lane batching (Section 6), UC1 takes 8.3–20.6 s over the 30 variants (before it, 13.8–26.4 s), 0.4–1.3 s (2–14%) more than UC2; the online phase takes 0.56–0.78 s. With the AdamW model UC1 classifies 64–72 of 100 images in single batch and 116–133 of 192 in multi-batch. Its accuracy first looked like chance level: a bisection over the added optimizations found the threaded GEMM (with the extra threads off, 5 of 10 correct) and, for unfused BN, the BatchNorm triples, both described in Section 7.

8.2 Final accuracy

Every variant stays above 60% (Fig. 15 and Table 16). The spread between variants (62–72 of 100, 118–131 of 192) is truncation noise: the variants draw different amounts of randomness, so different values wrap. The adders are exact, with 0 of 32,768 random ReLU inputs wrong for each adder. Fused and unfused BN are equally accurate with this model. With the older `Cifar_adam_001` model, probabilistic truncation in the 32-bit ring cost 10–17 points (Trio 56% against 73% in plaintext; 73% with a 64-bit ring). The AdamW model is much less sensitive to it.

Setting	images	correct
Plaintext, float (PyTorch)	100 / 192	76 / 138 (71.9%)
Trio 3PC, standard configuration	100	71
Trio 3PC, reduced-slack truncation (TRUNC_APPROACH=1)	100	70
2PC ABY2, single batch, all 30 variants	100	62–72
2PC ABY2, multi-batch, all 30 variants	192	118–131
2PC ABY2, A_UNKNOWN=0, single / multi-batch	100 / 192	64–72 / 116–133
2PC ABY2, public weights, single / multi-batch	100 / 192	63–73 / 115–131
2PC ABY2, CUT on / off (plain family, 12 runs each)	1,752	1,134 / 1,138
2PC ABY2 A2bits, TS{L} / TS_Mix / TS1, $F = 5$ (Table 21)	256	159–164 / 157–164 / 156–161
2PC ABY2 A2bits, TS{L} / TS_Mix / TS1, $F = 8$	256	139–169 / 186–190 / 186–187
Plaintext, float (PyTorch), the same 256 images	256	189

Table 16: Final accuracy with the AdamW-trained ResNet50 on the first CIFAR-10 test images.

9 The all_opt configurations on ImageNet

The artifact’s single-batch 2PC configurations (`measurements/configs/artifacts/triad/2pc/single_batch`) combine the online optimizations of the protocol: six configurations, A2bits and reshared, each in the three use cases of Section 2: UC1, weights known to none (`A_UNKNOWN=0`); UC2, weights known to the model owner (`A_UNKNOWN=1`, here also known in preprocessing, `MODELWEIGHTS_KNOWN_DURING_PREPROCESSING=1`); and UC3, weights known to all (public weights). Each runs ResNet50 on one ImageNet image with the three adders (`FUNCTION_IDENTIFIER=87, 187, 287`) and with `COMPRESS=0, 1`: 36 builds. As in the configurations, the model and the image are dummies (`MODELOWNER=-1, DATAOWNER=-1`); the runtime does not depend on their values.

Builds. *As given:* the configuration unchanged (`hpmc 161cf1d`); its conv triples use CHEETAH’s `HomConv2DSS`. *Optimized:* the configuration plus every optimization of this report (final code: `hpmc 92a2c0e`, `ConvTriple 8d8f244`, `flexNN 417f12b`; rounds 1–3 at `hpmc 1421df0`): `CHEETAH_CONV_PACKED=1` with the pipeline, `CUT_FRACTIONAL_BITS_OPT=1`, `ADDITIONAL_RELU_THREADS=24` (equal to the configuration’s `ADDITIONAL_GEMM_THREADS`, as the two share one pool), 100,000-word buffers, and `A2B_CONV_BAKE=1` for the A2bits configurations; everything else of Sections 3 to 5 and of the rounds below is on by default (the communication-optimized output repacking of round 4 is a separate variant); Table 17 lists every switch. `RNG_AHEAD=1`, part of the optimized set until this last round, is left out: at the final code it saves the Zen 4 hosts about 0.04s of online time (median over the 36 builds) but costs the Zen 3 hosts up to 0.17s (UC3, `COMPRESS=1`: 0.51 against 0.36s); a profile puts that in the online draws, most likely waiting for the ring’s producer thread. Each build runs three times per machine pair (median): the as-given builds alternating with the final code, the optimized builds in a separate series.

Correctness. The same flags on CIFAR-10 with the AdamW model and real data (P0 / P1 as owners, 10 images) classify 3–7 images correctly for every configuration with `COMPRESS=0` at the final code (plaintext: 6), including all UC2 builds. The outputs change only where a change alters the randomness that the probabilistic truncation sees (the bake’s privacy fixes of rounds 4 and 7, P1’s prescribed shares in UC2 A2bits in round 6); every other change leaves all 18 output hashes as they were. `COMPRESS=1` evaluates the ReLU on bits 12–19 only (`REDUCED_BITLENGTH_m=12, k=20`). With `FRACTIONAL=5` the carry dropped from the 12 low bits flips the sign of activations below 128.0, and even the plain RCA build without any of our changes classifies 0 of 10. It is a runtime setting only, and we keep it running rather than accurate. Its A2bits builds aborted with the element-parallel levels: the A2B bake committed conv masks for 8 of the 32 slices, so most outputs drew fresh masks. The bake now requires a full-width ReLU, and `COMPRESS=1` runs the A2B unbaked.

The A2bits configurations *as given* are not correct. They set `A2B_ONLINE_OPT=1`, which precomputes the A2B’s secret operand $[c] = \text{bool}(-\lambda)$ by a Boolean addition in preprocessing, but not the bake (`A2B_CONV_BAKE`), and without it the preprocessing pass generates the MSB adder’s Beaver triples for the Boolean addition’s input while the online phase uses its output $[c]$. On CIFAR-10 with the AdamW model, the as-given A2bits flags classify 0–2 of 10 images in all three use cases and every adder, as do the optimized flags without the bake (same output hashes as as-given in UC3); with the bake they classify 3–6. The reshared configurations do not use `A2B_ONLINE_OPT` and classify 4–7 of 10 as given.

Why the bake remarks ReLU inputs. A ReLU on $m = v + \lambda$ runs its MSB adder on $s_1 = \text{bool}(m)$, public, and $s_2 = \text{bool}(-\lambda)$, secret, since $\text{bool}(m) \boxplus \text{bool}(-\lambda) = \text{bool}(v)$. `A2B_ONLINE_OPT` computes $[c] = \text{bool}(-\lambda)$ in preprocessing, from the parties’ $\text{bool}(-\lambda_0)$ and $\text{bool}(-\lambda_1)$ by a Boolean addition (31 levels of AND gates). Two facts fix *when* this can happen. The adder’s Beaver triples are defined by the masks of its input wires,

[c] among them, and are requested while the preprocessing pass walks the circuit. And that pass is a single non-interactive sweep that fixes masks, sends one-way messages and records triple requests; the triples are generated in bulk afterwards. So [c] cannot be computed inside the pass, which would need 31 interactive rounds mid-sweep. It cannot be computed after the pass either, which is the as-given configuration: its triples then belong to a different mask. It must exist *before* the pass, and with it every ReLU input's mask λ .

The bake therefore commits the masks first. Each party i fixes a mask ℓ_i per ReLU input before the pass, the Boolean addition runs on $\text{bool}(-\ell_i)$, and the pass then has to deliver every ReLU input under the mask $\ell = \ell_0 + \ell_1$. How depends on the layer that produces the input:

- **A convolution or FC layer with secret weights** (UC1, UC2) ends with a message that re-masks its output under a mask each party draws: the party draws ℓ_i instead. With weights known in preprocessing, P1's triple share is chosen so that its truncated output mask is ℓ_1 ; such masks lie in the SecureML truncation's image (top FRACTIONAL bits zero), so P1 commits ℓ_1 there, uniformly. In the CHEETAH ResNet50 layout 8.00 of the 9.01 million ReLU inputs come straight from a convolution (with BatchNorm fused into it).
- **A residual sum** (0.80 million inputs): the addend computed last draws $\ell_i - \lambda'_i$, where λ' is the other addend's mask, so that the sum carries ℓ (round 5). P1's masks in UC2 must lie in the image, which $\ell_1 - \lambda'_1$ does not for a free λ'_1 ; so P1 commits the other addend's mask m_b as well (a convolution draws it as a PRF value in the image, a ReLU's outputs are committed anyway) and commits $\ell_1 = m_a + m_b$ for the sum, exactly the mask the local addition gives; the partner then draws m_a (round 6).
- **A convolution with public weights** (UC3) is local, so its output mask is $W * \lambda_{\text{in}}$, a linear function of the previous bit injections' masks; choosing those so that $W * \lambda_{\text{in}} = \ell$ is a linear system over $\mathbb{Z}_{2^{32}}$ without a solution in general. But every ReLU input's mask is a function of the input's masks and the earlier bit-injection masks alone, so the pass first runs the network over the masks, with each ReLU recording its input masks and outputting committed bit-injection masks, then runs the Boolean addition on the actual masks, and then the real sweep (round 4). With secret weights the same mask-only forward lets the conv triples start before the pass (round 5).
- **Every other producer** fixes λ without such a choice: in UC1 and UC2 the stem's pooling, whose BatchNorm passes its output on (0.20 million inputs). Such an input is moved onto the committed mask: party i sends $\delta_i = \ell_i - \lambda_i$ in preprocessing, and online both parties add $\delta_0 + \delta_1$ to m , which gives $v + \ell$. Since ℓ_i is uniform and known only to party i , δ_i is a one-time pad of λ_i ; the price is one ring element per value and party, in preprocessing only: 1.5 MiB.

Each committed mask must mask exactly one value, or the public masked values and remask messages reveal differences of secret values or masks; and a mask may lie in the truncation's image only where nothing the other party sees is masked by it alone (UC2, where P0 forms the masked value from its own share). Rounds 4, 6 and 7 below fixed four violations of the first version. In preprocessing every ReLU input that its producer moved is checked against its committed mask. In the first version, UC1 and UC2 remasked 1.00 million inputs (7.7 MiB) and UC3 all 9.01 million ($9.01 \cdot 10^6 \cdot 2 \cdot 4$ bytes = 69 MiB, exactly what the round-3 builds sent more than the as-given ones); now UC1 and UC2 remask 0.20 million and UC3 none.

What the bake costs. The reference is the same build with `A2B_CONV_BAKE=0`: the as-given A2B, which runs the same Boolean addition (after the pass) and computes wrong signs, but sends what an A2B without the bake sends. At the final code on flare / polynize (three interleaved runs per build, all nine A2bits builds with `COMPRESS=0`; with `COMPRESS=1` the bake is off):

- **Communication.** The triple generation and the online phase send the same bytes with and without the bake, in every build. The preprocessing pass sends 0.80 MB more in each direction in UC1 and UC2, the remask of the stem's 200,704 ReLU inputs (1.53 MiB, 0.1% of their 1.0–1.8 GiB of preprocessing), and nothing more in UC3. So the bake is free in communication with public weights, and costs the stem's remask with secret weights; free there too would need the stem's pooling truncation (or its BatchNorm) to commit its output mask (Section 11).
- **Preprocessing time.** With secret weights -0.10 to $+0.09$ s (UC1 0.05–0.10 s less, UC2 0.03–0.09 s more): the bake moves the Boolean addition from after the pass into the OT phase, next to the conv triples, and adds the commitment (and in UC2 P1's residual commitment). With public weights 0.06–0.16 s more: the mask-only forward, about 0.3 s of serial work, partly hidden.
- **Online time.** 0.00–0.07 s *less*: the baked A2B reads the committed [c] directly, where the unbaked one retrieves the Boolean addition's outputs. A first measurement (round 6) had put the bake at up to 0.1 s more online and 0.2 s more preprocessing: the baked A2B prepared its values serially, and the commitment was derived on one thread; round 7 removed both.

Where the time went. In a profile at `hpmc aeab3ae` (before the last two changes below), preprocessing was bound by OT generation: in a CPU profile over all threads, ferret’s LPN step took 31–44%, the MITCCRH key schedules 8–11% and MPCOT 10–13%. Per A2bits RCA inference the OT packs’ setup cost 0.9s, the bake’s Boolean addition 1.5s (31 carry levels over 9.0 million values; one level pays a ferret extension of all 32 packs), the conv triples 0.7s and the network’s preprocessing pass 0.5s; PPA4 added 3.1s of Beaver tuples. 84% of the online CPU samples were idle workers: RCA’s 1,714 rounds at about 0.3ms each are close to its floor, and disabling the NIC’s interrupt coalescing, which cut the ping from 0.25 to 0.08ms, shortened the RCA online phase by only 7%. Table 19 shows the final code.

Changes.

- **Worker wake-ups.** Between two circuit levels a worker spun about 100µs and then slept on a futex, shorter than a round trip, so every level woke 24 threads. Workers now wait about 20ms, on the counter’s own cache line: with the counter next to the completion count, every completion invalidated the line for all spinning workers, which slowed the Zen 3 hosts’ UC3 builds by up to 35%. On AMD CPUs the wait uses `MONITORX/MWAITX`, which parks the thread on the cache line without taking issue slots from its SMT sibling.
- **Vectorized GEMM accumulation** (`A_KNOWN=1`). The online GEMM called a share-level kernel per multiply-accumulate on $\{m, \lambda\}$ structs, about 0.2 MAC per cycle. The kernel of each variant is a sum of plain ring products, so it now runs as $C = XY^T$ on `uint32` matrices with a register-blocked 4×32 kernel (AVX-512 or AVX2), with the weight operand cached across a layer’s calls. The results are bit-identical; with weights known in preprocessing, P1’s accumulation, which its mask/send never reads, is skipped. The conv layers’ online time falls by a quarter on Zen 3; on Zen 4 the accumulation was not on the critical path.
- **The A2B bake** is restricted to full-width ReLUs (above).
- **Conv triples alongside the preprocessing pass** (`CHEETAH_CONV_ASYNC`). The conv triples used to start after the pass, whose single thread fixes their input masks. Now the batched HE pipeline starts on its own thread together with the pass and takes each layer as soon as the pass has recorded its masks; the rest of the preprocessing waits for it before the next generator uses the channels. Only the layers the pass reaches last leave a tail.
- **Parallel preprocessing levels** (`STREAM_PARALLEL_PRE`). The A2B, adder and bit-injection levels of the pass run on the worker pool like online. The pass’s append-only streams (triple types and triple inputs, the pre-send buffer, stored outputs) get per-worker cursors, reads use the online cursors, and every worker’s use of each stream is checked against the first element’s. The GEMM levels stay serial. Both changes leave all 18 CIFAR-10 output hashes and two multi-batch hashes unchanged.

More HE/OT threads help the OT-heavy builds: `CHEETAH_THREADS=64` instead of the configurations’ 32 (64 OT packs) shortens preprocessing by 8% (median) on Zen 4 and 4% on Zen 3, by up to 20% for PPA4, but lengthens it by up to 30% for the UC3 RCA builds, where setting up 64 packs costs more than it saves. Two ferret threads per pack, busy-polling the receive path and 31 instead of 24 workers did not help.

Results. Figure 16 shows all 36 builds; Table 18 summarizes them. At the final code every build is faster than as given online, on both pairs, and in preprocessing all but two UC3 builds with `COMPRESS=1` on Zen 3, which take 1–2% longer (1.88 against 1.86s and 2.01 against 1.97s): rounds 4–7 did not change their code, and the difference is within the run-to-run spread of these 2-second runs. In UC1 and UC2 the packed, pipelined and concurrent conv triples with the exact wire format (since round 5 alongside the OT phase) cut preprocessing from 11.7–17.3s to 1.7–5.5s on the Zen 4 pair ($2.9\text{--}7.1\times$) and from 15.1–21.9s to 2.7–6.9s on the Zen 3 pair ($3.0\text{--}6.0\times$). The online phase falls from 0.77–1.55s to 0.34–0.63s ($2.0\text{--}3.0\times$; Zen 3: 0.72–1.73 to 0.24–0.63s, $2.2\text{--}4.2\times$). In UC2, with the weights known in preprocessing, the online phase takes 0.34–0.56s on Zen 4 and 0.24–0.48s on Zen 3.

UC3 needs no linear-layer triples, so its preprocessing is OT-bound and changes least: $1.1\text{--}1.2\times$ faster ($0.98\text{--}1.1\times$ on Zen 3), with the online phase $1.5\text{--}2.2\times$ faster ($1.4\text{--}2.2\times$ on Zen 3). Before the parallel preprocessing levels, the Zen 3 hosts’ A2bits builds with `COMPRESS=1` preprocessed 0.11–0.14s *slower* than as given (five interleaved runs each). A one-flag-off ablation placed all of it in `ADDITIONAL_RELU_THREADS`, whose per-call stream-cursor checks slowed P0’s last preprocessing step from 0.32–0.34 to 0.44–0.49s. Unchecked versions for the preprocessing share removed that but slowed the `COMPRESS=0` builds by as much, a code-layout effect we did not keep. The parallel preprocessing levels now more than make up for it. As-given A2bits also skips the A2B bake and is not correct (Correctness).

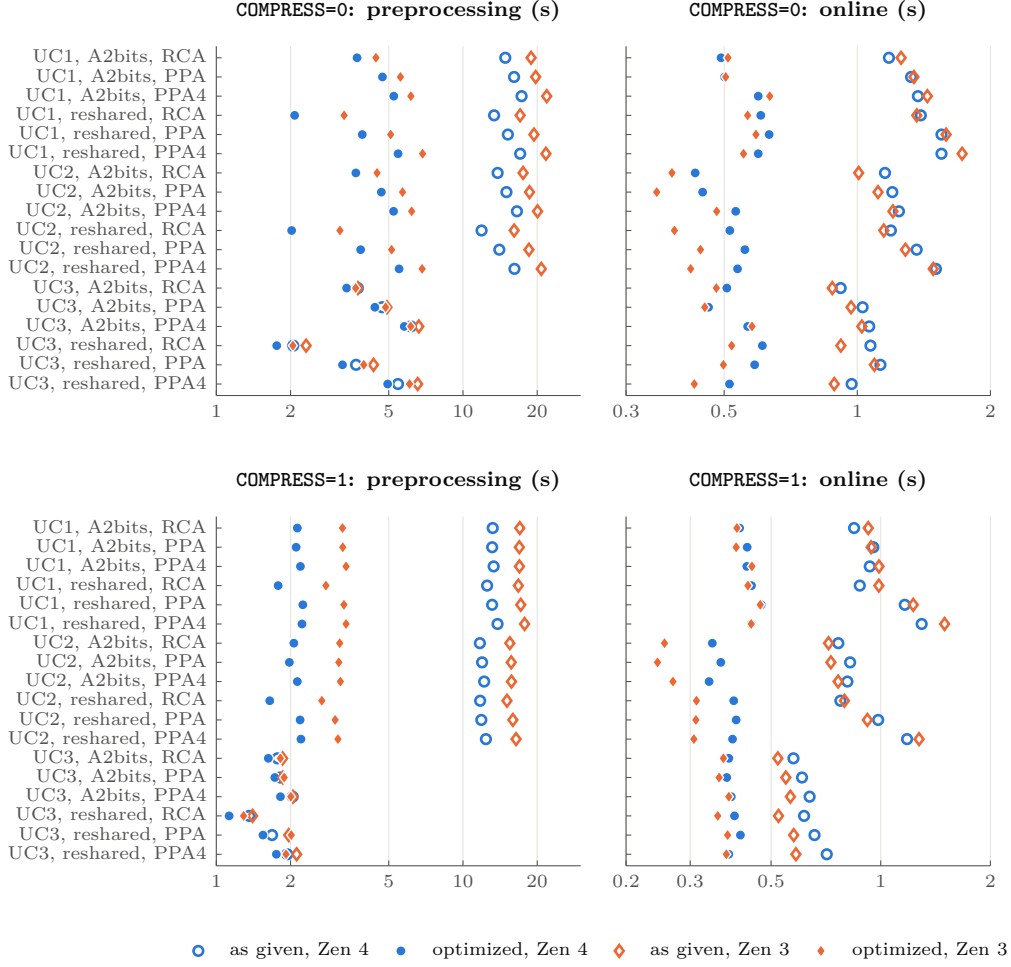


Figure 16: The 36 all_opt builds on ImageNet (dummy weights) by use case, as given and optimized, on both machine pairs (log scale; median of three runs).

Communication. Figure 17 shows the traffic of the same builds as P0 counts it; P1’s counters mirror P0’s, and the reruns and the Zen 3 pair send the same bytes. In UC1 and UC2 the optimized builds send $1.35\text{--}1.86\times$ less in preprocessing with **COMPRESS=0** (1,323–3,225 to 790–2,150 MiB) and $1.61\text{--}1.92\times$ less with **COMPRESS=1** (1,219–2,353 to 696–1,308 MiB). The conv triples account for nearly all of it: in UC2 they fall from about 1,000 to 479 MiB (Table 3), in UC1, whose AB triples need both cross products, from 2,003 to 958 MiB. With the weights known in preprocessing, both UC2 builds also send P1’s share corrections, 4 bytes per conv output (42 MiB). The OT-based tuples are unchanged, except that with **CUT_FRACTIONAL_BITS_OPT** PPA4 needs fewer multi-input AND tuples: 12 MiB less for resharded and 74 MiB less for A2bits, where the bake lets the cut skip the b -product gates. The network’s preprocessing pass changes by -67 to $+2$ MiB in UC1 and UC2 (of the bake’s remark only the stem’s ReLU inputs are left, 1.5 MiB: the residual sum’s 6.1 MiB are baked since rounds 5 and 6). UC3 needs no linear-layer triples, and its traffic is unchanged. In round 3 its A2bits builds with **COMPRESS=0** sent 69 MiB more than as given: no ReLU input comes straight from a masked conv layer, so the bake rebased every one of the 9.0 million inputs onto its committed mask, which the as-given builds skip (and compute wrong signs, see Correctness). The mask-only forward of round 4 bakes them without that message, so the final builds send exactly what the as-given ones do, and compute correctly. Online, **CUT_FRACTIONAL_BITS_OPT** removes 4–8% of the traffic of the UC1 and UC2 builds with **COMPRESS=0** (A2bits PPA4: none). The cut needs **TRUNC_DELAYED=0**, so it does not apply to UC3; with **COMPRESS=1** the online traffic does not change.

Round 4. Four further changes; the measurements in rounds 4–6 are from flare / polynize, the rerun of all 36 builds at the final code (Tables 18 and 19, Figs. 16 and 17) from both pairs.

1. **P1’s bake in the pass.** The conv triples of resharded PPA4 did not run at half speed next to the pass: they waited for P1, whose pass took 1.28–1.43 s against P0’s 0.38–0.50 s. P1 bakes the reshare bits into every conv output mask in the (serial) GEMM level, and with **CUT_FRACTIONAL_BITS_OPT** the PPA4

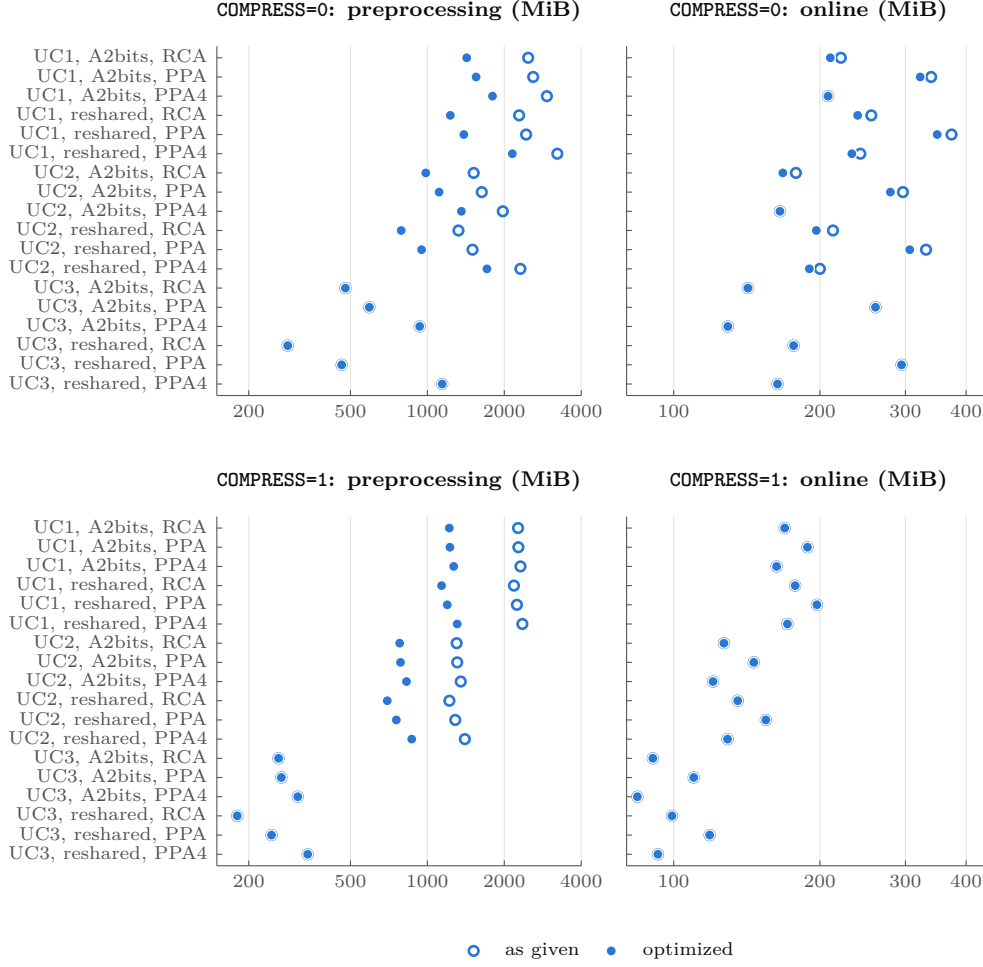


Figure 17: Traffic of the 36 all_opt builds on ImageNet (dummy weights) by use case, as given and optimized: MiB sent plus received by P0, i.e. in both directions (log scale). Preprocessing counts the triple generation (HE, OT, key exchange) and the network’s preprocessing pass. Every run and both machine pairs send the same bytes.

slice ranks were recomputed for each of the 32 slices of each of the 7.3 million outputs. With the ranks fixed at compile time P1’s pass takes 0.55–0.66 s, and reshared PPA4 preprocesses in 5.5–5.7 s instead of 6.0–6.4 s (UC1 and UC2), with the same outputs bit for bit.

2. **Privacy of the bake.** The committed masks are addressed by the next A2B’s position, and every convolution took them. So the downsample convolution and the one after it masked their outputs with the same masks (1.5 million values: the differences of secret activations were public), and at the residual ReLU the remask message $\ell_i - \lambda_i$ gave away the downsample convolution’s mask, since ℓ_i had also masked its partner. And the committed masks replayed the generator stream the passes then drew their own masks from. Now only a convolution feeding a baked ReLU takes committed masks, the others draw fresh ones, and the committed values come from a separate stream. Neither change costs traffic or time; the results of rounds 1–3 stand.
3. **UC3 bake for free.** The mask-only forward described above: UC3 A2bits sends 69 MiB less in preprocessing (546 to 478 MiB for RCA, 1,002 to 933 for PPA4, –7 to –13%) in the same time (3.52–5.82 against 3.54–5.92 s), and its online phase gets slightly faster (0.56–0.64 against 0.63–0.68 s), since no ReLU input is moved. In preprocessing every ReLU input’s mask is checked against the mask-only forward’s; none of the 49 ReLUs differs.
4. **Output repacking,** the communication-optimized variant (CHEETAH_CONV_REPACK=1, Section 3.1): 285 instead of 479 MiB per HE product, preprocessing traffic –18 to –31% in UC1 (1,229–2,150 to 845–1,766 MiB) and –11 to –24% in UC2 (790–1,710 to 598–1,518 MiB), for 1.3–1.7 s more preprocessing; the online phase is unchanged.

Round 5. Two more changes.

1. **Conv triples alongside the OT phase.** During the OT phase only about 20 of the 64 hardware threads are busy (sampled every 50 ms), while the conv triples ran with the pass and after it. Every ReLU's outputs now take committed masks (values of a PRF under the party's key, by position), so every convolution's input masks are known before the pass: the pass first runs the network over the masks, where each convolution records its triple's inputs and stops, then starts the conv triples on four channels of their own, runs the OT phase (moved into the pass), and then the real sweep, which checks every convolution's inputs against the recorded ones. With all 32 threads the conv triples slowed the OT phase down by about as much as they gained, with 16 throughout they became the tail of the Zen 3 builds with short OT phases (up to 0.7 s slower); they now take 16 while the OT phase runs and all 32 after it. Against round 4 on Zen 4 preprocessing takes 0.03–0.34 s less over the 12 UC1 and UC2 builds; over all 36 builds, the round-5 code preprocesses a median 0.15 s faster than round 3 on Zen 4 (up to 0.8 s) and 0.03 s on Zen 3, where a few builds with short OT phases are up to 0.11 s slower. The online phase gets 0.02–0.07 s faster, since the bit injections no longer draw masks.
2. **Residual sums baked.** The layer computed last of a residual sum's two addends draws $\ell - \lambda'$, where λ' is the other addend's mask, so that the sum carries ℓ . In UC1 this removes the residual sum's remask (6.1 MiB); in UC2 P1's masks must lie in the SecureML truncation's image, so P1 still sent its half alone (3.1 MiB saved; round 6 removes the other half). The outputs are unchanged bit for bit. In preprocessing, every baked ReLU input's mask is checked against its committed mask; the check showed that with dummy weights P1 holds a bias mask that the UC2 bake cannot compensate, so P1's A2B inputs were off after biased convolutions in the dummy-weight ImageNet runs (the timing measurements are unaffected).

Round 6. In UC2, P1 now also bakes the residual sum. P1's partner cannot draw $\ell_1 - \lambda'_1$ for a free λ'_1 , since with weights known in preprocessing P1's conv masks are the truncation's image of its prescribed triple share. So the other addend's P1 mask is committed as well: a convolution producing it draws a PRF value in the image (uniform there, like a fresh mask, so the prescribed share stays uniform), a ReLU producing it has committed bit-injection masks anyway (round 5). P1's committed mask of the sum is then $\ell_1 = m_a + m_b$, exactly the mask the local addition gives, and the partner draws m_a , again in the image. Which sums are committed follows from the network alone, so both parties agree on who sends what. This removes the remaining 3.1 MiB (ImageNet, every UC2 A2bits build with COMPRESS=0: 3.21 MB fewer received by P0, nothing else changes); on the CIFAR ResNet50, whose every block has a residual sum (4 produced by a convolution, 12 by a ReLU), the UC2 pass traffic falls from 16.8 to 12.3 MB, and P1's check of every baked ReLU input passes with the real weights. Convolutions now cost the A2B no message at all: in UC1 and UC2 only the stem's pooling output is remasked.

The round also fixes a leak in UC1: the committed masks were narrowed to the truncation's image in every build with TRUNC_DELAYED=0, but in UC1 P1's masks are free draws that hide the truncated share it sends, so a narrowed mask revealed part of it (about one bit per value on average); only builds with weights known in preprocessing narrow now. UC1's outputs do not depend on these masks and stay bit for bit the same. Neither change costs time: against round 5, the 36 builds preprocess a median 0.01 s slower on Zen 4 (−0.21 to +0.10 s) and as fast on Zen 3 (−0.12 to +0.16 s), and the online phase stays within 0.04 s. (An A/B series right after the change had put UC1's A2bits RCA build 0.1–0.2 s behind round 5, all of it in P1's OT setup; in the final series the identical binary ran 0.05–0.1 s *faster* than round 5, so that was series-to-series variation, not the change.)

Round 7. The bake's runtime and one more privacy fix.

1. **Repeated committed masks.** The counter-mode PRF behind the committed masks (the mask-only forward's truncation masks since round 4, every committed ReLU output mask since round 5, UC2's residual masks of round 6) took its block index and slot as $k/\text{BUFFER_SIZE}$ and $k \bmod \text{BUFFER_SIZE}$ of a macro without parentheses: $k/512/32$ and $(k \bmod 512)/32$. Each value came back for 32 consecutive masks, 16 distinct ones per 16,384, so each committed mask hid many values and the other party saw differences of secret activations. Masks never change the work, so no timing, traffic or accuracy number was affected; it showed when the commitment below drew from the same PRF and P1's narrowed UC2 masks came out tiny (SecureML truncation wraps, 0–2 of 10). The macro is parenthesized, and the PRF checks its outputs before the committed masks are drawn. The CIFAR outputs of all builds with committed masks change; all 18 classify 3–7 of 10.
2. **Baked A2B on the pool.** The baked A2B read $[c]$ through one global cursor, so its prepare and complete levels ran serially, where the unbaked A2B runs them on the worker pool. Each value now reads its slices by index.
3. **Parallel commitment.** The committed masks come from the PRF by slot, on 16 threads, in the OT phase.

Against round 6, preprocessing is unchanged (median ± 0.00 s on Zen 4, -0.01 s on Zen 3) and the online phase a median 0.01 s and up to 0.10 s faster.

Round 8. The bake’s Boolean addition, a ripple-carry adder of the two parties’ private inputs with one AND (two random OTs, one per direction) per bit and value, took 1.44–1.52 s of the A2bits builds’ preprocessing on flare / polynize. A per-round log showed where: each of its 31 dependent rounds generated its own random OTs, split per OT worker into chunks of about 250,000 values, each with its own round trip, and every few rounds the ferret instances ran dry and a whole extension round fell into the dependent chain.

1. **Random OTs first (A2B_ADDER_BATCH).** The random OTs do not depend on the inputs, so all rounds’ OTs are generated before the first round, in one call per OT pack and direction, starting next to the masks’ commitment; the extension round then runs at full parallelism and outside the chain. A round only exchanges its two correction bits per AND (in four messages) and computes the shares; the adder’s loops run on bit-position-contiguous arrays (the value-major layout had a stride of 32 words and took longer than the exchanges). The 31 rounds now take 0.1 s together.
2. **The cut in the adder (A2B_ADDER_CUT).** Under CUT_FRACTIONAL_BITS_OPT every A2B replaces the top FRACTIONAL slices of $[c]$ by the constant 0, so the Boolean addition stops after bit $\ell - F - 1$: 26 instead of 31 AND rounds and 16% fewer OTs. It applies only if every conversion of the network takes the cut, which the INIT pass records; in these networks all A2B inputs are ReLU inputs.

At the same code with the two flags off and on (median of two runs, Zen 4), the nine COMPRESS=0 A2bits builds preprocess 0.45–0.75 s faster (3.19–5.14 instead of 3.46–5.73 s), the COMPRESS=1 ones 0.05–0.12 s, and the online phase does not change. The Boolean addition is now its OT volume: 1.04–1.21 s, of which about 1.0 s are the random OTs (one extension round of all 64 ferret instances and the hashing) and 0.1 s the rounds. All A2bits builds still classify 4–7 of 10 CIFAR-10 images, and multi-batch (192 images) 134 (MODELWEIGHTS_KNOWN=0) and 124 (=1). These are Zen 4 numbers only (algofi / goracle were booked).

Round 9. The generated a-known circuits. The circuit generator now emits the a-known four-way circuit correctly at any width (Section 10), so A2bits PPA4 takes it again at the cut, where it had taken the AB circuit since CUT_NARROW_32 (A2BITS_PPA4_AB=1 keeps that), and two changes make the generated a-known circuits cheaper online.

1. **One mask share per dot group.** The members of a dot group are pending: only the XOR of their output-mask shares, the root’s mask, is ever used. All members but one leaf of the root’s XOR chain now take mask 0 and that leaf the root’s mask. The 27-bit a-known four-way adder draws 56 instead of 197 random values, which the parallel ReLU constructors take from a stream filled serially; the a-known circuit’s online phase was about 0.1 s slower than the AB circuit’s, now 0.02–0.03 s (0.05 s at 64 bits in UC1).
2. **The folded LSB of the a-known RCA.** Its first carry, the a-known product $x_{k-1}y_{k-1}$, had to be remasked before the next AND (one round and one message). Now $c_{k-2} = x_{k-2}y_{k-2} \oplus x_{k-2}x_{k-1}y_{k-1} \oplus x_{k-1}(y_{k-2}y_{k-1})$ is one dot group with the same triple. At 32 bits the A2bits RCA under the bake therefore takes the narrow cut adder as well (the identity-substituted circuit would need its hand edits redone): 49 fewer rounds in every use case (UC2 1,665 instead of 1,714; 64 bits 2,890 instead of 2,939) and 2.2 MiB less online traffic.

On ImageNet (interleaved, median of 3), A2bits PPA4 with the a-known narrow adder against the AB one in the same run preprocesses 368 MiB less (UC1–UC3: 1,699 / 1,280 / 779 against 2,068 / 1,648 / 1,147 MiB) in 1.0–1.2 s less time (4.50 / 4.49 / 4.31 against 5.69 / 5.53 / 5.34 s), for 0.02–0.03 s more online time (0.438 / 0.361 / 0.417 against 0.412 / 0.339 / 0.389 s) with the same online traffic and rounds; at 64 bits 795 MiB and 2.8–3.1 s less (UC2 2,631 against 3,426 MiB, 8.31 against 11.08 s; online 0.610 against 0.585 s). The A2bits RCA builds against CUT_NARROW_32=0 (the old identity cut) in the same run: online 0.468 / 0.390 / 0.436 against 0.481 / 0.423 / 0.506 s, preprocessing unchanged (+2 MiB).

Reduced-slack truncation. ABY2 now runs the reduced-slack truncation TS1 of Truncation Untangled [3] (its Fig. 9; Fig. 18) fused into the ReLU: TRUNC_APPROACH=4 (TS_Mix: TS1 in the ReLUs, the probabilistic TS{L} elsewhere) and TRUNC_APPROACH=1 (TS1 also for the poolings, whose division joins the next ReLU’s TS1), both with TRUNC_DELAYED=1 and the A2B bake. In 3PC the party that holds the mask computes TS1’s mask-side values; in ABY2 nobody does, but their inputs are the carries of the two mask shares’ sum, which the bake’s Boolean addition already computes. Two narrow COTs per value make the carries arithmetic, and a narrow multiplexer gives the bit injection its one extra product; the public side (the masked value’s top bits and its MSB) is local, so the truncation sends nothing online.

1. **The cut in the Boolean addition.** Literally, TS1 needs the carry into the top bit, so the Boolean addition would run full width. Instead it adds the shares shifted locally by F (a pooling’s $1/d$ folded in) over $\ell - F$ bits, which keeps round 8’s cut (26 instead of 31 rounds); the A2B converts the truncated value with the cut as well, and TS1 runs in the ring $2^{\ell-F}$. The dropped low carries are compensated by rounding the public side: the result is off by -1 , 0 or $+1$ from the floor, with mean $+0.03$ LSB from the real value. The exact low carry (TS1_LOW_CARRY, 0 or $+1$ as in 3PC) costs a full-width addition and a 31-bit COT per value; only the poolings’ ReLUs take it, whose small averages need it.
2. **The cut for UC3’s TS{L} (A2B_DELAYED_CUT).** With public weights TS{L} delays the truncation to the ReLU, whose A2B therefore converted the untruncated value, without the cut. The same shifted addition gives it the cut: 245 fewer online rounds and 10.8 MiB less online traffic for RCA, and 21–110 MiB less preprocessing traffic. The reshared builds have no bake; there each party shifts its own additive share of the A2B input by F (their sum is the truncated value minus 0 or 1): RCA 245 fewer rounds and 16.1 MiB less online, PPA 24.7 MiB, PPA4 9.6 MiB. With the cut PPA4 computes 30–46 ms longer online in every build, although it sends less; preprocessing and online together are still shorter.

Table 20 shows the ImageNet costs. TS_Mix and TS1 need the bake, so they apply to the nine A2bits configs without COMPRESS; the others keep TS{L}. TS_Mix sends the same online and 36–39 MiB more in preprocessing (the first version, with a full-width addition, 57–58 MiB), 0.07–0.36 s more preprocessing time; TS1 removes the last pooling’s message. Online, all three take the same time, once the bit injection computes TS1’s public side from the A2B’s input instead of storing it per value (that cost 0.02–0.04 s). Since TS1 has no wrap failures, FRACTIONAL can grow, and the cut adder shrinks with it: TS1 with $F = 8$ runs 148 fewer online rounds than TS{L} with $F = 5$ (RCA; PPA 13.6 MiB less online and 0.5 s less preprocessing) and classifies like the plaintext model on CIFAR-10 (Table 21), where TS{L} loses 20–50 images of 256 at $F = 8$ and collapses at $F = 10$. (TS_Mix divides a pooling before a ReLU, as in the ImageNet ResNet’s stem, probabilistically at scale 2^{3F} , which wraps at $F \geq 8$; TS1 folds it into the ReLU.) At $F = 5$ all variants classify 156–166 of 256, the fixed point’s precision being the limit.

Exact truncation. ABY2 had no exact truncation: the generic TE0 (A2B, shift, B2A) and TE1 (Fantastic Four’s carry correction) need B2A and share-level truncation primitives it does not implement. Both now run fused into the ReLU on TS1’s machinery (TRUNC_APPROACH=2 / =3, TRUNC_DELAYED=1, the A2B bake), exact and without a message of their own. With $z = m + \nu$ and $a = m \bmod 2^F$, TS1’s full design (with the low carry w_t of the mask shares and the offset $2^{\ell-1}$) yields $\lfloor z/2^F \rfloor + [a \neq 0] - c_t$ for $z \geq 0$, where c_t is the carry into bit F of $m + \nu$ and $[a \neq 0]$ is public. Only $z \geq 0$ matters, since DReLU zeroes the rest, so the wrap of $m + \nu$ is linear there and, unlike in 3PC, no slack is needed. c_t is the MSB of an $(F + 1)$ -bit a-known adder of $(0, a)$ and $(0, \nu \bmod 2^F)$, that is of $\text{bool}(m)$ and the bake’s $[c]$, which runs in the same rounds as the ReLU’s adder (TE_LOW_ADDER: the folded RCA for RCA builds, $F - 1$ messages; the a-known four-way PPA for PPA and PPA4 builds, 4 messages in 2 rounds at $F = 5$, against the Sklansky PPA’s 10 in 5); with $c_t = m_c \oplus \lambda_c$ it enters the bit injection linearly, from $[\lambda_c]$ and $[\lambda_b \lambda_c]$, the B2As of λ_c and of the Boolean product $\lambda_b \lambda_c$ (whose two cross terms take a COT of one bit each): 2ℓ bits per value instead of $3\ell - 1$ with a full-width multiplexer for the product. Figures 18 to 20 state the three protocols. The two differ in DReLU: TE1 takes it of $A = \lfloor m/2^F \rfloor + \lfloor \nu/2^F \rfloor = \lfloor z/2^F \rfloor - c_t$ with the cut (the two values differ only where the output is 0 either way), TE0 of z at full width, exact for every z . Online, TE1 costs TS1 plus the low adder’s messages; TE0 adds the rounds the cut saves. Every positive input of the unit test (4,096 random values, $|v|$ up to 2^{31} for TE0 and 2^{30} for TE1) comes out exact, with all three adders at 32 and 64 bits. On CIFAR-10, TE0 and TE1 compute the same values and classify 158 ($F = 5$) and 185 ($F = 8$) of the 256 images in all three use cases; exact truncation floors (mean error -0.23 LSB), which costs nothing at $F = 8$ (TS1 187, plaintext 189).

Table 22 shows the ImageNet costs at the final code. TE1’s online phase takes about as long as TS1’s (0.36–0.46 against 0.35–0.49 s) for 8.4 MiB more online traffic with every adder (the low adder’s messages; 10.5 / 21 MiB with the earlier RCA and Sklansky low adders) and the same rounds with RCA (PPA4 +3, PPA +35); its preprocessing grows by 0.1–1.2 s and 140 MiB (RCA), 158 MiB (PPA) or 216 MiB (PPA4, whose low adder takes a Beaver 3-tuple): the full-width Boolean addition, the $(\ell - 1)$ -bit COTs of w_t , λ_c and $\lambda_b \lambda_c$, the low adder. Against the previous version (four-way low adder, B2A products, a-known narrow PPA4, folded RCA) TE1 sends 4.3 MiB (RCA) to 12.6 MiB (PPA, PPA4) less online, PPA4 56 rounds fewer, and preprocesses 18 MiB (RCA) and 373 MiB (PPA4) less. TE0’s full-width DReLU costs 288 more rounds with RCA and up to 0.12 s of online time (PPA4).

9.1 TS1, TE1 and TE0 in ABY2: protocols and security

This subsection states the truncations of Section 9 as protocols in the style of Truncation Untangled [3], proves the identities they compute and argues their security. All three run fused into a ReLU: the input is a delayed

Protocol Π_{TS1} (shifted design, fused into the ReLU)

Input: $\langle z \rangle = (m, [-\nu])$; a public factor f with $t' \geq 0$ extra fractional bits ($f = 1, t' = 0$ for a plain ReLU, $f \approx 2^{F+t'}/d$ for a folded d -element average). $\ell' = \ell - F, K' = 2^{\ell'-1-t'}$.

Output: $\langle \text{DReLU}(y_1) \cdot y \rangle$ with $y = \text{round}(fz/2^{F+t'}) + e, e \in \{-1, 0, 1\}$.

Offline:

1. $P_i: \rho_i = ((f\nu_i) \bmod 2^\ell) \gg F$, reduced mod $2^{\ell'}$.
2. $\mathcal{F}_{\text{ADD}}^B(\rho_0, \rho_1)$ over ℓ' bits: $[\rho[j]]^B$ for all j (the A2B's Boolean mask input), the carry $w = \gamma_{\ell'-1}, r_{\text{msb}} = \rho[\ell' - 1]$ and, if $t' > 0, w_t = \gamma_{t'}$.
3. $\text{B2A}_{F+1+t'}(w), \text{B2A}_{F+1+t'}(r_{\text{msb}})$ (both only enter times K'), and if $t' > 0$ $\text{B2A}_{\ell-1}(w_t)$; P_i sets $[l_a]_i = -((\rho_i \bmod 2^{\ell'-1}) \gg t') + K'[w]_i - [w_t]_i$ ($[w_t]_i = 0$ for $t' = 0$).
4. The preprocessing of $\Pi_{\text{MSB}}^{\ell'}$ with the DReLU output mask $\lambda_b: [\lambda_b], [\lambda_b l_a]$ (the bit injection's usual product) and $[\lambda_b r_{\text{msb}}]$ by $\mathcal{F}_{\text{MUX}}^{F+1+t'}$.

Online (no message of its own):

1. $M_0 = ((fm + 2^{F-1}) \gg F) + 1 \bmod 2^{\ell'}; \langle b \rangle^B = \neg \Pi_{\text{MSB}}^{\ell'}(M_0, [\rho]^B)$, the DReLU of $y_1 = M_0 + \rho \bmod 2^{\ell'}$ (the cut adder).
2. $c = -M_0 + 2^{\ell'-2} \bmod 2^{\ell'}, s = 1 - 2c[\ell' - 1], c' = (c \bmod 2^{\ell'-1}) \gg t', M = (2^{\ell'-2} \gg t') - c' - K'c[\ell' - 1]$.
3. $T = M - \lambda_T$ with $[\lambda_T] = [l_a] + sK'[r_{\text{msb}}]$ and $[\lambda_b \lambda_T] = [\lambda_b l_a] + sK'[\lambda_b r_{\text{msb}}]$; output $\Pi_{\text{BI}}(\langle b \rangle^B, T)$.

Figure 18: TS1 in ABY2 (TRUNC_APPROACH=1 / =4, ts1_generate_tuples, ts1_lift_shift, prepare_opt_bit_injection_ts1). The full design (TS1_LOW_CARRY) runs the same steps in the full ring: $\rho = \nu$ over ℓ bits, $M_0 = m$, offset $2^{\ell-1}, t' = F$ and $w_t = \gamma_F$; it is the first half of Π_{TE1} .

convolution output z at scale 2^{2F} , the output $\text{DReLU}(z)$ times the truncated value, and neither sends a message beyond those of the ReLU's MSB adders and its bit injection.

Notation. Values live in $\mathbb{Z}_{2^\ell}$ with F fractional bits; $K = 2^{\ell-1-F}$. An ABY2 share of x is $\langle x \rangle = (m_x, [\lambda_x])$: the masked value m_x is public to both parties, the mask $\lambda_x = \lambda_x^0 + \lambda_x^1$ is additively shared (P_i holds λ_x^i), and $x = m_x - \lambda_x$. For the ReLU input we write $\nu = -\lambda_z = \nu_0 + \nu_1$ ($\nu_i = -\lambda_z^i$, the shares of the code's nu), so $z = m + \nu$. A Boolean share is $\langle b \rangle^B = (m_b, [\lambda_b]^B)$ with $b = m_b \oplus \lambda_b$; $[u]$ denotes additive shares of a bit u in $\mathbb{Z}_{2^\ell}$, $[u]^B$ XOR shares, $x[j]$ bit j of $x, x \gg j$ the logical shift and $\llbracket P \rrbracket \in \{0, 1\}$ the indicator of P . The protocols call

- $\mathcal{F}_{\text{COT}}^k$: P0 inputs $x_0 \in \mathbb{Z}_{2^k}$, P1 a bit x_1 ; the outputs satisfy $c_0 + c_1 = x_0 x_1 \bmod 2^k$ (ferret's silent COT, Section 4.1). $\text{B2A}_k(u)$ turns $[u]^B$ into $[u] \bmod 2^{k+1}$ with one call: P_i outputs $u_i - 2c_i$, since $u_0 \oplus u_1 = u_0 + u_1 - 2u_0 u_1$.
- $\mathcal{F}_{\text{MUX}}^k$: from $[b]^B$ and $[r]$, outputs $[br] \bmod 2^k$ (two COTs, one per direction).
- $\mathcal{F}_{\text{ADD}}^B$ (the A2B bake's Boolean addition): P0 inputs x_0 , P1 inputs x_1 , both of k bits; outputs XOR shares of every bit of $x_0 + x_1 \bmod 2^k$. Since bit j of the sum is $x_0[j] \oplus x_1[j] \oplus \gamma_j$, each party's share of the carry γ_j into bit j is its share of the sum bit XOR its own input bit, without communication. The implementation is an RCA with one COT multiplication per AND gate.
- $\Pi_{\text{MSB}}^k(p, [x]^B)$: the a-known ROT-preprocessed MSB adder of width k (RCA, PPA or four-way PPA, Section 4.1) with a public summand p and a Boolean-shared one; outputs $\langle \text{MSB}(p + x \bmod 2^k) \rangle^B$. Its preprocessing (triples, tuples, output masks) does not depend on the inputs; online it opens the masked value of every gate output, each under a fresh mask.
- $\Pi_{\text{BI}}(\langle b \rangle^B, T)$: the bit injection of a bit into a value $T = M_T - \lambda_T$ given as a public part M_T and $[\lambda_T]$, with $[\lambda_b]$ and $[\lambda_b \lambda_T]$ preprocessed. Locally $[bT] = m_b M_T - m_b [\lambda_T] + (2m_b - 1)([\lambda_b \lambda_T] - M_T [\lambda_b])$ (from $b = m_b + (1 - 2m_b)\lambda_b$); then each party adds its share of a fresh output mask and sends the sum, so both learn m_{bT} : one ℓ -bit message per party, the ReLU's own message.

Correctness of Π_{TS1} . **Lemma 1.** If y_1 , read as a signed ℓ' -bit value, lies in $[-2^{\ell'-2}, 2^{\ell'-2})$, then $T = \lfloor y_1/2^{t'} \rfloor$ for $t' = 0$ and $T - \lfloor y_1/2^{t'} \rfloor \in \{0, 1\}$ for $t' > 0$; and $y_1 = \text{round}(fz/2^F) + 1 - E$ with $E \in \{0, 1, 2\}$.

Proof. The second claim: $fz = fm + f\nu_0 + f\nu_1 \bmod 2^\ell$, and shifting the three summands by F separately loses the carries out of their low F bits, $E \in \{0, 1, 2\}$ in all (the rounding constant moves into M_0 , the $+1$ centres the error). The first is TS1's identity [3, Fig. 9] in the ring $2^{\ell'}$ on $x' = 2^{\ell'-2} - y_1 \in (0, 2^{\ell'-1}]$, whose masked value is $c = x' + \rho$ and whose mask ρ has the two shares ρ_0, ρ_1 ; it is the computation of Lemma 2 below

Protocol Π_{TE1} (exact truncation fused into the ReLU)

Input: $\langle z \rangle = (m, [-\nu])$ with $z \in [-2^{\ell-1} + 2^F, 2^{\ell-1})$. *Output:* $\langle \text{DReLU}(z) \cdot \lfloor z/2^F \rfloor \rangle$, exact.

Offline:

1. $\mathcal{F}_{\text{ADD}}^B(\nu_0, \nu_1)$ over all ℓ bits: $[\nu[j]]^B$ and the carries; $w = \gamma_{\ell-1}$, $w_t = \gamma_F$, $r_{\text{msb}} = \nu[\ell-1]$.
2. $\text{B2A}_{F+1}(w)$, $\text{B2A}_{F+1}(r_{\text{msb}})$, $\text{B2A}_{\ell-1}(w_t)$; P_i sets $[l_a]_i = -(a_i \gg F) + K[w]_i - [w_t]_i$ with $a_i = \nu_i \bmod 2^{\ell-1}$.
3. The preprocessing of $\Pi_{\text{MSB}}^{\ell-F}$ (the DReLU, output mask λ_b) and of the low adder Π_{MSB}^{F+1} (output mask λ_c).
4. $[\lambda_c] = \text{B2A}_{\ell-1}(\lambda_c)$. $d = \lambda_b \lambda_c$: two $\mathcal{F}_{\text{COT}}^1$ calls give XOR shares of the cross terms $\lambda_b^0 \lambda_c^1$ (P0 inputs λ_b^0 , P1 λ_c^1) and $\lambda_c^0 \lambda_b^1$; P_i XORs them with $\lambda_b^i \lambda_c^i$; $[\lambda_b \lambda_c] = \text{B2A}_{\ell-1}(d)$.
5. $[\lambda_b]$, $[\lambda_b l_a]$ as for every ReLU; $[\lambda_b r_{\text{msb}}]$ by $\mathcal{F}_{\text{MUX}}^{F+1}$.

Online:

1. $a = m \bmod 2^F$. In the same rounds: $\langle b \rangle^B = \neg \Pi_{\text{MSB}}^{\ell-F}(m \gg F, [\nu[F..\ell-1]]^B)$, the DReLU of $A = (m \gg F) + (\nu \gg F) \bmod 2^{\ell-F}$ (the narrow adder on the top slices of the conversion), and $\langle c_t \rangle^B = (m_c, [\lambda_c]^B) = \Pi_{\text{MSB}}^{F+1}(0 \| a, 0 \| [\nu[0..F-1]]^B)$, the carry into bit F of $m + \nu$.
2. $c = -m + 2^{\ell-1}$, $s = 1 - 2c[\ell-1]$, $c' = (c \gg F) \bmod 2^{\ell-1-F}$, $M = K - c' - Kc[\ell-1]$; $\llbracket a \neq 0 \rrbracket = (a + 2^F - 1) \gg F$; $s_c = 1 - 2m_c$.
3. $T = M_T - \lambda_T$ with $M_T = M - \llbracket a \neq 0 \rrbracket + m_c$, $[\lambda_T] = [l_a] + sK[r_{\text{msb}}] - s_c[\lambda_c]$ and $[\lambda_b \lambda_T] = [\lambda_b l_a] + sK[\lambda_b r_{\text{msb}}] - s_c[\lambda_b \lambda_c]$; output $\Pi_{\text{BI}}(\langle b \rangle^B, T)$.

Figure 19: TE1 in ABY2 (TRUNC_APPROACH=3; ts1_generate_tuples, te_generate_products, get_msb_range with g_te_narrow_top, prepare_opt_bit_injection_ts1). The low adder is the a-known RCA for RCA builds and the a-known four-way PPA otherwise (TE_LOW_ADDER).

Protocol Π_{TE0} (exact truncation fused into the ReLU, exact DReLU)

Input: $\langle z \rangle = (m, [-\nu])$, any z . *Output:* $\langle \text{DReLU}(z) \cdot \lfloor z/2^F \rfloor \rangle$, exact. As Π_{TE1} , except:

- Offline 3: the preprocessing of the full-width Π_{MSB}^{ℓ} instead of $\Pi_{\text{MSB}}^{\ell-F}$.
- Online 1: $\langle b \rangle^B = \neg \Pi_{\text{MSB}}^{\ell}(m, [\nu]^B)$, the DReLU of z itself; the low adder reads the low F slices of the same conversion and runs in its first rounds.

Figure 20: TE0 in ABY2 (TRUNC_APPROACH=2). It pays the rounds and gates the cut saves (RCA: F more AND levels).

with ℓ' for ℓ , t' for F and offset $2^{\ell'-2}$. Without w_t the lift is exact for $t' = 0$ because $\lfloor y_1/2^0 \rfloor = y_1$ has no low bits to carry out of. $\square$

A simulation of the formulas over 10^6 random inputs ($\ell = 32$, $F = 5$, $f \in \{1, 4\}$; $f = 28$, $t' = 3$; $\ell = 64$, $F = 12$) finds the lift exact (or $\{0, 1\}$) everywhere and $y - \text{round}(fz/2^{F+t'})$ in $\{-1, 0, 1\}$; the unit test RELU_TS1 (4,096 inputs, all adders, 32 and 64 bits) agrees.

Correctness of Π_{TE1} and Π_{TE0} . **Lemma 2.** For $z \in [0, 2^{\ell-1})$ and $\lambda_y = l_a + sKr_{\text{msb}}$: $M - \lambda_y = \lfloor z/2^F \rfloor + \llbracket a \neq 0 \rrbracket - c_t$. Hence $T = M_T - \lambda_T = \lfloor z/2^F \rfloor$.

Proof. Let $x' = 2^{\ell-1} - z \in (0, 2^{\ell-1}]$; then $c = -m + 2^{\ell-1} = x' + \nu \bmod 2^{\ell}$. Split $c = 2^{\ell-1}c_h + c_l$ and $\nu = 2^{\ell-1}r_{\text{msb}} + \nu_l$ with $c_l, \nu_l < 2^{\ell-1}$. As $x' + \nu_l < 2^{\ell}$, its bit $\ell-1$ is $c_h \oplus r_{\text{msb}}$, so over the integers $x' = c_l - \nu_l + 2^{\ell-1}(c_h + r_{\text{msb}} - 2c_h r_{\text{msb}}) = c_l - \nu_l + 2^{\ell-1}c_h + 2^{\ell-1}s r_{\text{msb}}$. Dividing by 2^F , $\lfloor x'/2^F \rfloor = \lfloor c_l/2^F \rfloor - \lfloor \nu_l/2^F \rfloor - \beta + Kc_h + sKr_{\text{msb}}$ with the borrow $\beta = \llbracket c \bmod 2^F < \nu \bmod 2^F \rrbracket$. The mask's low part is $\nu_l = a_0 + a_1 - 2^{\ell-1}w$, so $\lfloor \nu_l/2^F \rfloor = (a_0 \gg F) + (a_1 \gg F) + w_t - Kw$. Substituting $M = K - \lfloor c_l/2^F \rfloor - Kc_h$ and l_a gives $M - \lambda_y = K - \lfloor x'/2^F \rfloor - \beta$, and since $\lfloor z/2^F \rfloor = K - \lfloor x'/2^F \rfloor$, $M - \lambda_y = \lfloor z/2^F \rfloor + \llbracket x' \bmod 2^F \neq 0 \rrbracket - \beta$. Finally let $\zeta = z \bmod 2^F$ and $n = \nu \bmod 2^F$: $x' \bmod 2^F \neq 0$ iff $\zeta \neq 0$; $c \bmod 2^F = (n - \zeta) \bmod 2^F$, so $\beta = \llbracket \zeta \neq 0 \wedge n \geq \zeta \rrbracket$; and $a = (\zeta - n) \bmod 2^F$ with $a + n = \zeta + 2^F c_t$. The cases $\zeta = 0$, $n < \zeta$, $n = \zeta$ and $n > \zeta$ show that $\llbracket a \neq 0 \rrbracket - c_t$ is 1 exactly when $\zeta \neq 0$ and $n < \zeta$, and 0 otherwise, as is $\llbracket \zeta \neq 0 \rrbracket - \beta$. With $c_t = m_c + s_c \lambda_c$, $M_T - \lambda_T = M - \lambda_y - \llbracket a \neq 0 \rrbracket + c_t$. $\square$

Lemma 3. With $b = \neg \text{MSB}_{\ell-F}(A)$ as in Π_{TE1} , $b \cdot T = \text{DReLU}(z) \lfloor z/2^F \rfloor$ for $z \in [-2^{\ell-1} + 2^F, 2^{\ell-1})$.

Proof. $m + \nu = z$ gives $A = \lfloor z/2^F \rfloor - c_t \bmod 2^{\ell-F}$, and K is half that ring. For $z \geq 0$, $\lfloor z/2^F \rfloor \in [0, K)$, so $A \in [-1, K)$: $A = -1$ only if $\lfloor z/2^F \rfloor = 0$, where $T = 0$ (Lemma 2) and the product is 0 for either bit; otherwise $b = 1$. For $z < 0$ in the range, $\lfloor z/2^F \rfloor \in [-K + 1, -1]$, so $A \in [-K, -1]$ and $b = 0$, whatever T is. In Π_{TE0} , $b = \text{DReLU}(z)$ exactly. $\square$

Lemma 2 was also checked exhaustively for $\ell = 8$, $F \in \{2, 3\}$ (all $z \geq 0$ and all ν_0, ν_1) and by simulation for $\ell \in \{16, 32, 64\}$; the unit tests RELU_TE1 / RELU_TE0 find every positive input exact with all three adders at 32 and 64 bits.

Security. Theorem 2. Π_{TS1} , Π_{TE1} and Π_{TE0} securely compute their outputs against a semi-honest adversary corrupting one party, in the hybrid model with $\mathcal{F}_{COT}$, $\mathcal{F}_{MUX}$, $\mathcal{F}_{ADD}^B$ and the preprocessing of Π_{MSB} .

Proof sketch. Offline. Every value a party receives is an output of an ideal functionality, which hands it a share that is uniform on its own (the other party’s share completes it); the local steps (shifts, l_a , the carries’ shares) are functions of its view. The simulator draws these shares uniformly. *Online.* The messages are the opened masked values of the gates of the MSB adders (one adder in Π_{TS1} , two in Π_{TE1} and Π_{TE0}) and the bit injection’s m_{bT} . Each is the sum of a value and a fresh mask of which the corrupted party holds one share and the other share is uniform and used nowhere else: the gates’ output masks are drawn independently in preprocessing, λ_c and λ_b are independent of each other and of ν (the cross terms of d go through $\mathcal{F}_{COT}$), and the bit injection’s output mask is fresh. So every opened value is uniform given the rest of the view, and the simulator sends uniform values (for m_{bT} , the masked output it derives from the corrupted party’s output share and mask share). The public quantities M_0 , c , s , M , a , $\llbracket a \neq 0 \rrbracket$ and m_c are functions of opened values. The bake’s committed masks are as in the rest of the A2bits protocol (one value per slot, Section 9). $\square$

What is left. Table 19 splits the time of six builds at the final code; the ranges below are over all 18 builds with COMPRESS=0. With secret weights, preprocessing is now the OT phase (1.4–4.7s, inside the pass), the pass’s own work (0.33–0.58s, including its mask-only forward), the steps after it (0.22–0.56s) and 0.1–0.3s of key exchange, input preparation and gaps; the conv triples run alongside the OT phase and finish before the pass does.

1. **OT generation** remains the largest cost. Every build pays 0.84–1.04s to set up the 32 ferret packs (about 0.1s more with secret weights, whose conv triples run alongside); the tuples after that take 0.26–0.34s for RCA but 1.6–3.6s for PPA4, whose Beaver 3- and 4-tuples need up to six more extension rounds of all 32 packs. In the round-3 CPU profile, ferret’s LPN step took 33–41% of all samples, the AES key schedules of the correlation-robust hash 8–11%. The last extension round is only partly used, but that wastes at most one round (about 0.3s for PPA4). A fixed-key hash would remove most of the key schedules, but that is a security decision (Section 11).
2. **The bake’s Boolean addition** cost the A2bits builds 1.41–1.53s: 31 dependent COT-multiplication rounds of 15–22ms plus an extension round. Since round 8 it is 1.04–1.21s, almost all of it the OTs it needs (two per AND); a log-depth adder would need fewer rounds but more AND gates, and so more OTs, and B2A of random bits halves the OTs but cannot give P1 the prescribed masks of UC2’s residual sums.
3. **The conv triples** no longer show: in round 3 the HE ran 0.28–0.60s past the pass (reshared PPA4: 1.0–1.1s, from P1’s serial bake, fixed in round 4). Running alongside the OT phase, they compete with it for cores and memory bandwidth, which the longer OT setup shows.
4. **Serial work** in the pass: the GEMM levels (the weights-known-in-preprocessing path pushes into vectors there) and the bake’s input preparation still run on one thread; with public weights the mask-only forward adds 0.3s to the A2bits pass (0.55–0.74 against 0.24–0.30s).
5. **Online**, RCA’s 1,714–2,012 rounds cost 0.16–0.19s of waiting. Of the rest, the main thread spent 0.11s handing levels to the pool, 0.05s transposing bits for the A2B serially, and 0.06s in layer glue (copies, residual additions, freeing buffers) in the round-3 profile.
6. **Communication:** output repacking (round 4) removes 40% of the conv-triple traffic at 1.3–1.7s of CPU; it pays off below about 1.2 Gbit/s (UC2) or 2.1 Gbit/s (UC1). Of the bake’s remask, only the stem’s pooling output is left (1.5 MiB, Section 11).

10 64-bit rings

With $\ell = 64$ the fixed-point slack is no concern (at $F = 12$ to 16 the truncations never wrap in practice), but the 2PC configurations did not run: the preprocessing had no 64-bit linear-layer triples, the ROT-preprocessing adders existed for $k = 8, 16, 32$ only, and every optimization of the A2B was written for 32 bits. All of it now works at $\ell = 64$ (hpmc acc5728, ConvTriple 2d0b448; docs/BITLENGTH64.md).

HE with $t = 2^{64}$. SEAL’s plaintext moduli have at most 60 bits, so the packed evaluator does BFV with $t = 2^{64}$ itself: encoding $\lfloor qm/2^{64} \rfloor$ exactly as $Dm + \lfloor rm/2^{64} \rfloor$ for $q = D \cdot 2^{64} + r$ (which removes BFV’s $(q \bmod t)$ error term), centered 64-bit weight shares, outputs switched to two primes, composed by CRT and truncated, and an own decryption $\lfloor x \cdot 2^{64}/q' \rfloor$. A product’s noise is a sum of $(e + \rho)w$ over up to $2^{19.4}$ weight terms per output coefficient (ResNet50’s widest layers), below 2^{78} except with probability 2^{-40} . The flooding ($\Delta/16 = 2^{96}$) keeps the 32-bit parameters’ margin of 19 bits above it, so q needs 165 bits (three 55-bit primes, $N = 8192$; the first version took 180 bits). An ImageNet HE product then sends 1,013 MiB, $2.1\times$ the 32-bit 479 MiB: the

query has $1.5\times$ the bits per coefficient, the response about $1.9\times$, and $N = 8192$ costs the sparse output layout its $\sqrt{N}$ factor (Table 4). Output repacking works at 64 bits as well (a 53-bit special prime, 218 bits in all, the limit for 128-bit security): 464 MiB, below the 32-bit default, for $2.8\times$ the HE time.

The adders. The circuit generator’s RCA and PPA builders produce any width; for the four-way PPA, which it builds by hand per width, a width-generic tree (groups of three slices, blocks of up to four per level) reproduces the 64-bit circuit and produces every other width. The a-known four-way circuit needed hand fixes at 32 bits (operand order, a-known products in dot chains that must stay pending with their own mask shares); the generator now emits them itself, and A2bits PPA4 takes the generated a-known circuit at 64 bits as well (Section 9): 795 MiB and 2.8–3.1 s less preprocessing than the AB circuit it took before, for 0–0.05 s more online time.

The cut as a narrow adder. At 32 bits the cut of the redundant top F slices identity-substitutes the full-width circuits by hand: public constants on the vacant slices, gate skipping with mask retargeting, and triple bookkeeping that the bake maps must mirror. At 64 bits the MSB adder is simply narrower: the A2B prepares the full width as before and an adder of width $64 - F$ runs on slices $F, \dots, 63$; its MSB is the sign of the value, correct by construction. The narrow adders of all eight MSB families are generated for $F \in \{8, 10, \dots, 20, 24\}$. At $F = 12$ the RCA needs 51 instead of 63 AND gates per value (52 instead of 64 rounds), PPA and PPA4 19–20% fewer triples, and the bake’s Boolean addition 51 instead of 63 rounds; A2B_DELAYED_CUT and TS1’s cut design follow unchanged.

The same construction exposed a 32-bit bug: under the A2B bake, the identity-substituted PPA and PPA4 circuits (AB and a-known alike) flip DReLU for inputs near the cut’s limit 2^{31-F} , at a rate that grows with $|v|$ (40–45% in the top octave; 114 of 4,096 random inputs at $F = 5$, all above 2^{16}), while RCA and the builds without the bake are correct. With CUT_NARROW_32 the 32-bit PPA and PPA4 under the bake now take narrow adders too, which fixes it. On ImageNet (A2bits, interleaved with CUT_NARROW_32=0) the narrow PPA needs 33 fewer rounds and 15 MiB less online traffic (online 0.39 against 0.41 s in UC2) for 0.1–0.25 s more preprocessing. PPA4 first traded the a-known circuit, which had no generated form, for the AB one, whose Beaver 3- and 4-tuples took 323 MiB more OT material; with the generated a-known narrow circuit it keeps the a-known one (Section 9). The a-known RCA, whose generated circuits fold the a-known LSB carry, takes the narrow adder as well: 49 fewer rounds per inference.

The reshare bake and the arithmetic triples. The reshare bake’s maps (which slice consumes which random multiplication and which 3-tuple) are read off the generated 64-bit and narrow circuits and checked by the generator script; the generator’s skipped input re-maskings become `zero_add_local`, which falls back to the real re-masking for inputs whose mask is not baked, as in the 32-bit files. At two narrow widths the generator skips re-maskings with 4-tuple fields that the bake cannot serve; there they stay. P1’s ReLU preprocessing traffic drops by 15% for PPA on CIFAR-10. One more 64-bit defect was silent: the elementwise arithmetic triples behind every secret-by-secret product (max pooling) came from gemini’s HE in SEAL’s plaintext space, and its bit-length assertion is compiled out in release builds, so with 64 bits they were random. Gilboa multiplication over the silent COTs fixed them first (for bit j of one factor, one COT of width $64 - j$); now gemini’s CRT-batched HE product itself takes 64-bit shares: the product of two shares plus a mask that hides it statistically (2^{-40}) needs $2 \cdot 64 + 1 + 40 = 169$ bits of CRT plaintext, five 34-bit primes, which keeps the plaintext scaling after the switch to the 49-bit prime. Its mask was also too wide (the top limb kept the complement bits: 87-bit masks at 32 bits, harmless but wasteful). For 409,600 AB2 triples the HE product takes 0.23 s and 57 MB against Gilboa’s 1.42 s and 129 MB (AB: 0.30 s, 114 MB against 2.05 s, 248 MB); ARITH_OT=1 keeps Gilboa.

Results. Table 23 compares the 18 all_opt builds with COMPRESS=0 at 64 bits ($F = 12$) with the same builds at 32 bits ($F = 5$), interleaved on flare / polynize. Preprocessing traffic grows by $1.7\text{--}2.1\times$ (the HE products $2.1\times$, the OT material $1.7\text{--}2.1\times$: the Boolean triples and tuples scale with the slices, the COTs’ messages with ℓ) and online traffic by $2.0\times$; preprocessing takes $1.5\text{--}2.0\times$ as long (PPA4 the most, whose tuples dominate), the online phase $1.6\text{--}2.0\times$. RCA pays the most rounds, 50 instead of 25 AND levels per ReLU (2,890 against 1,665 rounds in UC2, with the folded a-known RCA); the prefix adders add one level (49 more rounds). Repacking takes the HE products of UC2 A2bits RCA from 1,050 to 524 MiB (1,265 instead of 1,789 MiB in all, $1.37\times$ the 32-bit traffic) for 1.8 s more preprocessing. On CIFAR-10 (AdamW ResNet50, 256 images, UC2 A2bits RCA) the 64-bit builds classify like the plaintext model: TS{L} 188 at $F = 12$ and 189 at $F = 16$, TE0 and TE1 the same (32 bits: TS{L} 159 / 169 at $F = 5$ / 8; plaintext 189).

flag (optimized setting)	effect	where
<i>set in the optimized builds</i>		
CHEETAH_CONV_PACKED=1	conv triples with the packed HE evaluator (several images and output channels per ciphertext, lazy transforms, exact wire format, concurrent chunks) instead of CHEETAH's HomConv2DSS	Section 3
CHEETAH_CONV_PIPELINE=1	all convolutions in one pipelined call: encryption, evaluation and decryption of consecutive layers overlap	Section 3
CUT_FRACTIONAL_BITS_OPT=1	the MSB adder skips the bits the preceding SecureML truncation fixes (TRUNC_DELAYED=0): fewer AND tuples for PPA4, 4–8% less online traffic	Section 5
ADDITIONAL_RELU_THREADS=24	the elements of each ReLU level (MSB adder, A2B, bit injection) run on the worker pool of the GEMM threads	Section 5
SEND_BUFFER, RECV_BUFFER = 100,000 A2B_CONV_BAKE=1 (A2bits)	network buffers of 100,000 instead of 10,000 words every ReLU input's mask is committed before the preprocessing pass, which A2B_ONLINE_OPT needs to be correct	Section 5 below
<i>new defaults since the configurations (on)</i>		
GEMM_POOL_SPIN = 2 ²⁰ , GEMM_POOL_MWAITX=1	pool workers wait across a round trip on their own cache line, with MONITORX/MWAITX on AMD	Changes
GEMM_FAST=1	online GEMM of A_KNOWN=1 as a register-blocked uint32 matrix product	Changes
CHEETAH_CONV_ASYNC=1	conv triples alongside the preprocessing pass, each layer once its masks are known	Changes
STREAM_PARALLEL_PRE=1 CHEETAH_CONV_EARLY=1	the preprocessing pass's ReLU levels on the worker pool a mask-only forward records every conv's triple inputs; the conv triples run alongside the OT phase (CHEETAH_CONV_EARLY_THREADS: 16 of 32 threads during it, all after it)	Changes round 5
A2B_BAKE_MASK_PASS=1	UC3: the Boolean addition runs on the masks a mask-only forward computes, so no ReLU input is remasked	round 4
A2B_BAKE_RESIDUAL=1	residual sums carry the committed mask (in UC2, P1 also commits the other addend's mask)	rounds 5, 6
A2B_ADDER_BATCH=1	the bake's Boolean addition generates the random OTs of all its rounds first; a round only exchanges corrections	round 8
A2B_ADDER_CUT=1	under the cut the Boolean addition stops at bit $\ell - F - 1$: 26 instead of 31 rounds	round 8
<i>variants, off</i>		
CHEETAH_CONV_REPACK=1	communication-optimized conv triples (output repacking, $N = 8192$): −40% conv traffic, +1.3–1.7 s	round 4
A2B_BAKE_BN=1	a BatchNorm before a ReLU takes the committed masks (unfused BatchNorm only)	round 5
RNG_AHEAD=1	the party's own random stream produced ahead by a thread: −0.04 s online on Zen 4, up to +0.17 s on Zen 3	Builds
CHEETAH_THREADS=64	64 OT packs: −8% (median) preprocessing on Zen 4, up to +30% for UC3 RCA	Changes
TRIPLE_GPU (ConvTriple)	the packed pipeline's evaluator on a GPU, bit-identical: conv triples 2.1–2.3× faster	Section 3.3
GEMM_FAST_GPU=1	GEMM_FAST's online product on the GPU, exact; no gain at batch size one	Section 3.3

Table 17: The optimized ImageNet builds: flags set on top of the all_opt configurations, the defaults this work added after the configurations (hpmpc 161cf1d), and the variants left off. Everything else, e.g. the OT changes of Section 4 and TCP_NODELAY, was already in the code the configurations were given with.

builds	pair	preprocessing (s)		online (s)	
		as given	optimized	as given	optimized
UC1, known to none, COMPRESS=0 (6)	Zen 4	13.4–17.3	2.1–5.5	1.18–1.55	0.49–0.63
	Zen 3	17.1–21.9	3.3–6.9	1.26–1.73	0.50–0.63
UC1, known to none, COMPRESS=1 (6)	Zen 4	12.5–13.8	1.8–2.2	0.84–1.29	0.41–0.47
	Zen 3	16.8–17.8	2.8–3.4	0.92–1.50	0.40–0.47
UC2, known to one, COMPRESS=0 (6)	Zen 4	11.9–16.5	2.0–5.5	1.16–1.51	0.43–0.56
	Zen 3	16.1–20.8	3.2–6.8	1.01–1.49	0.35–0.48
UC2, known to one, COMPRESS=1 (6)	Zen 4	11.7–12.4	1.6–2.2	0.77–1.18	0.34–0.40
	Zen 3	15.1–16.4	2.7–3.2	0.72–1.27	0.24–0.31
UC3, known to all, COMPRESS=0 (6)	Zen 4	2.1–6.2	1.8–5.8	0.92–1.13	0.46–0.61
	Zen 3	2.3–6.6	2.0–6.1	0.88–1.09	0.43–0.58
UC3, known to all, COMPRESS=1 (6)	Zen 4	1.4–2.0	1.1–1.8	0.58–0.71	0.38–0.41
	Zen 3	1.4–2.1	1.3–2.0	0.52–0.58	0.36–0.38

Table 18: The all_opt builds on ImageNet by use case (A2bits and reshared, three adders each), as given and optimized at the final code (round 7; median of three runs). Per-build values: docs/TRIAD_ALL_OPT.md.

build	preprocessing (s)									online		
	total	OT setup	tuples	bake add.	pass	conv tail	after	other	(s)	rounds	wait (s)	
UC1, A2bits, RCA	3.62	0.99	0.26	1.47	0.37	0.00	0.22	0.31	0.50	1,818	0.16	
UC1, reshared, PPA4	5.48	0.99	3.60	0.00	0.38	0.00	0.40	0.11	0.60	779	0.14	
UC2, A2bits, RCA	3.72	0.99	0.26	1.53	0.33	0.00	0.28	0.33	0.48	1,714	0.19	
UC2, reshared, PPA4	5.47	1.00	3.55	0.00	0.39	0.00	0.44	0.10	0.53	675	0.14	
UC3, A2bits, RCA	3.58	0.95	0.29	1.53	0.55	0.00	0.16	0.10	0.53	1,963	0.17	
UC3, reshared, RCA	1.67	0.86	0.34	0.00	0.24	0.00	0.16	0.07	0.60	2,012	0.17	

Table 19: Where the time of six builds goes at the final code (round 7, COMPRESS=0, ImageNet, flare / polynize, phase timestamps, median of two runs). Tuples: OT-based triples and tuples after the OT setup; bake add.: the A2B bake’s Boolean addition; pass: the network’s preprocessing pass including its mask-only forward, without the OT phase that runs inside it (secret weights) or the Boolean addition (UC3 A2bits); conv tail: the part of the conv triples still running after the pass (since round 5 they run alongside the OT phase, and none is left); after: the rest of the preprocessing (COT, multiplexer, FC triples, triple bookkeeping); other: key exchange, the bake’s input preparation, starting the conv triples, and gaps. Wait: online time spent waiting for the other party.

UC	variant	pre MiB	pre s	online MiB	rounds	online s
UC1	TS{L}	1,341	3.18	210.2	1818	0.498
	TS_Mix	1,378	3.45	210.2	1818	0.453
	TS1	1,379	3.52	209.4	1817	0.465
	TS1, $F = 8$	1,377	3.44	202.9	1670	0.452
UC2	TS{L}	922	3.16	167.8	1714	0.421
	TS_Mix	959	3.36	167.8	1714	0.405
	TS1	960	3.45	167.0	1713	0.397
	TS1, $F = 8$	958	3.37	160.5	1566	0.387
UC3	TS{L}	421	2.94	125.3	1714	0.447
	TS_Mix	460	3.10	125.3	1714	0.427
	TS1	461	3.17	124.5	1713	0.452
	TS1, $F = 8$	459	3.15	118.1	1566	0.413
UC3 reshared	TS{L}	268	1.76	170.4	2008	0.550
	TS{L} + delayed cut	257	1.76	154.3	1763	0.501

Table 20: TS1 on ImageNet (the all_opt configs with RCA, A2bits unless marked, flare / polynize, dummy weights, median of 3 runs; P0’s traffic sent and received). TS{L}: the configs as given (TRUNC_DELAYED=0 in UC1 and UC2, =1 in UC3, with A2B_DELAYED_CUT).

F	UC1			UC2			UC3
	TS{L}	TS_Mix	TS1	TS{L}	TS_Mix	TS1	
5	164	164	161	159	164	161	162 / 157 / 156
8	167	190	187	169	186	187	139 / 188 / 186
10				44	188	189	

Table 21: Accuracy with TS1 (CIFAR-10, AdamW ResNet50, the first 256 test images; plaintext 189). With TS1_LOW_CARRY at $F = 5$: TS1 165 / 165 / 165, TS_Mix 158 / 157 / 159 (UC1 / UC2 / UC3).

build	variant	pre MiB	pre s	online MiB	online s	rounds
UC1 RCA	TS{L}	1,344	3.16	207.9	0.468	1769
	TS1	1,382	3.55	207.1	0.488	1768
	TE1	1,521	3.84	215.5	0.463	1768
	TE0	1,559	4.30	228.1	0.514	2056
UC1 PPA	TS{L}	1,443	4.28	306.7	0.471	949
	TS1	1,481	4.61	305.9	0.453	948
	TE1	1,639	4.93	314.3	0.443	983
	TE0	1,660	5.01	345.8	0.495	1021
UC1 PPA4	TS{L}	1,699	4.50	201.6	0.438	730
	TS1	1,737	4.74	200.8	0.417	729
	TE1	1,954	5.89	209.2	0.426	732
	TE0	1,989	6.23	215.5	0.528	732
UC2 RCA	TS{L}	924	3.16	165.6	0.390	1665
	TS1	962	3.42	164.8	0.404	1664
	TE1	1,102	3.78	173.2	0.412	1664
	TE0	1,139	4.24	185.8	0.426	1952
UC2 PPA	TS{L}	1,023	4.16	264.4	0.375	845
	TS1	1,061	4.51	263.6	0.360	844
	TE1	1,219	4.69	272.0	0.422	879
	TE0	1,240	4.87	303.5	0.433	917
UC2 PPA4	TS{L}	1,280	4.49	159.2	0.361	626
	TS1	1,318	4.73	158.4	0.353	625
	TE1	1,534	5.86	166.8	0.358	628
	TE0	1,570	6.10	173.1	0.477	628
UC3 RCA	TS{L}	423	2.97	123.2	0.436	1665
	TS1	463	3.23	122.4	0.423	1664
	TE1	602	3.47	130.7	0.464	1664
	TE0	640	3.89	143.4	0.483	1952
UC3 PPA	TS{L}	523	3.93	222.0	0.425	845
	TS1	562	4.36	221.3	0.400	844
	TE1	720	4.43	229.6	0.424	879
	TE0	741	4.55	261.1	0.474	917
UC3 PPA4	TS{L}	779	4.31	116.7	0.417	626
	TS1	818	4.71	116.0	0.376	625
	TE1	1,035	5.55	124.4	0.410	628
	TE0	1,071	5.86	130.7	0.499	628

Table 22: Exact truncation on ImageNet (A2bits, $F = 5$, flare / polynize, dummy weights, median of 3 interleaved runs; P0's traffic sent and received; RCA / PPA hpmc 8c2adac, PPA4 a058c36). TS{L}: the builds as given (TRUNC_DELAYED=0 in UC1 and UC2, =1 in UC3); all with the narrow cut adders.

build	pre MiB	pre s	online MiB	online s	rounds
UC1 A2bits RCA	1,344 / 2,682	3.16 / 4.89	208 / 416	0.47 / 0.87	1769 / 2994
UC1 A2bits PPA	1,443 / 2,891	4.28 / 6.64	307 / 620	0.47 / 0.78	949 / 998
UC1 A2bits PPA4	1,699 / 3,522	4.50 / 8.24	202 / 403	0.44 / 0.70	730 / 779
UC2 A2bits RCA	924 / 1,791	3.16 / 4.87	166 / 331	0.39 / 0.78	1665 / 2890
UC2 A2bits PPA	1,023 / 2,000	4.16 / 6.66	264 / 535	0.38 / 0.68	845 / 894
UC2 A2bits PPA4	1,280 / 2,631	4.49 / 8.31	159 / 318	0.36 / 0.61	626 / 675
UC3 A2bits RCA	423 / 738	2.97 / 4.49	123 / 246	0.44 / 0.81	1665 / 2890
UC3 A2bits PPA	523 / 947	3.93 / 6.06	222 / 450	0.43 / 0.72	845 / 894
UC3 A2bits PPA4	779 / 1,578	4.31 / 7.68	117 / 234	0.42 / 0.70	626 / 675
UC1 reshared RCA	1,175 / 2,409	2.13 / 3.65	239 / 474	0.52 / 1.00	1867 / 3092
UC1 reshared PPA	1,330 / 2,627	3.97 / 6.04	349 / 676	0.59 / 0.95	1004 / 1047
UC1 reshared PPA4	2,059 / 4,047	5.57 / 9.16	233 / 459	0.54 / 0.88	779 / 828
UC2 reshared RCA	756 / 1,518	2.13 / 3.30	197 / 389	0.46 / 0.88	1763 / 2988
UC2 reshared PPA	911 / 1,735	3.99 / 5.94	306 / 591	0.49 / 0.78	900 / 943
UC2 reshared PPA4	1,639 / 3,156	5.50 / 9.14	190 / 374	0.45 / 0.74	675 / 724
UC3 reshared RCA	257 / 469	1.71 / 2.81	154 / 304	0.51 / 0.92	1763 / 2988
UC3 reshared PPA	436 / 734	3.24 / 4.87	264 / 506	0.49 / 0.86	900 / 943
UC3 reshared PPA4	1,062 / 1,992	4.91 / 8.22	148 / 289	0.48 / 0.83	675 / 724

Table 23: The all_opt builds at 32 / 64 bits ($F = 5 / 12$; ImageNet, dummy weights, flare / polynize, median of 3 interleaved runs; P0’s traffic sent and received). COMPRESS=0. A2bits rows: hmpc 8c2adac (RCA, PPA) and a058c36 (PPA4), with the narrow cut adders (32-bit RCA too) and the generated a-known PPA4 at both widths; reshared rows: 4c2e446.

11 Limitations and open problems

- **Truncation.** ABY2 runs TS1, TE0 and TE1 only fused into a ReLU (Section 9); a stand-alone one (a pooling without a ReLU after it) would need a message of its own, and the reshared and COMPRESS=1 builds, which have no A2B bake, a Boolean addition of their own. A pooling’s division folded into a ReLU takes TS1’s shifted design, also under TE.
- **64-bit rings.** The a-known four-way adder has no generated 64-bit or narrow form, so A2bits PPA4 takes the AB circuit there (and, with CUT_NARROW_32, for the 32-bit cut under the bake); the narrow cut adders exist for the listed values of F only (Section 10). An HE product costs $2.1\times$ the 32-bit traffic without repacking.
- **Weights known in preprocessing.** The multi-batch a2b bake still excludes MODELWEIGHTS_KNOWN_DURING_PREPROCESSING, whose prescribed shares assume one word per register.
- **Ciphertext occupancy.** At 7×7 feature maps, inputs are 5–19% data and outputs 6–24%. Packing the sparse outputs (automorphisms, $N = 8192$ with a special prime) is implemented as the communication-optimized variant CHEETAH_CONV_REPACK (round 4 in Section 9): 285 instead of 479 MiB per HE product for 1.3–1.7s more preprocessing, worth it below about 1–2 Gbit/s. Packing the inputs remains open.
- **A2B bake.** UC1 and UC2 still remask the 0.20 million ReLU inputs after the stem’s pooling (1.5 MiB in preprocessing). Committing the pooling’s truncation mask, or a BatchNorm’s output mask like a convolution’s, would remove that; a BatchNorm bake works mechanically but changed the outputs of an unfused multi-batch check for reasons not yet understood, so it is off.
- **OT extension.** On the CPU, ferret’s LPN step and the AES key schedules of the correlation-robust hash are the largest costs of the Boolean preprocessing; a fixed-key hash is a security decision. On a GPU both run there (Section 3.3); what remains on the CPU is the packs’ setup (base OTs, IKNP, first extension), the latency-bound Boolean addition of the A2bits builds, and the few consumers that need the COT blocks themselves. The online phase measured 0.04–0.05s higher with the GPU OT path in the UC1 builds on the shared workstation, for reasons not yet found; a two-machine GPU setup would settle it. In multi-batch the OT kernels of both parties saturate the one GPU.
- **Multi-batch on one workstation.** With both parties on one machine, 62 GB hold two GPU processes per party (16 ImageNet images) and one CPU process per party, so the 24-process configuration of the artifact cannot run there. The online activations scale linearly with the lanes.

12 Conclusion

The largest runtime gains came from doing less redundant work in the HE evaluator (packing, lazy transforms, pipelining). The largest communication gains came from sending only information-bearing bits: the exact wire format, full BatchNorm ciphertexts, and OTs whose message bits carry several products. In the online phase, latency rather than bandwidth dominates, so the gains came from Nagle-free sockets, fewer rounds and parallelizing the per-round work at batch size one. Seeded, bit-reproducible runs made every change measurable and exposed the bugs in Section 7. Moving the A2B to preprocessing (Section 9) needs every ReLU input’s mask fixed before the preprocessing pass; most producers can commit their masks for free, and the care it takes is privacy bookkeeping: each committed mask may mask one value only, and a mask narrowed for a truncation only where the other party never sees what it hides.

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